Chapter 1: Introduction

Object Space vs. Image Space

Space	Description
Object Space	Continuous abstract space where shapes are defined using coordinates
Image Space	Discrete pixel grid (display monitor) where the image is produced

Key Areas

- **Geometric Representation** — defining objects using coordinates (vertices, etc.)
- **Transformation** — mapping objects to image space using matrices (controls position, orientation, size)
- **Scan Conversion** — converting continuous figures to discrete pixels (rasterization)
- **Anti-aliasing** — reducing the "staircase" distortion caused by scan conversion

2D Graphics Pipeline

$$\text{Object Coordinates} \rightarrow \text{World Coordinates} \rightarrow \text{Device Coordinates}$$

3D Graphics Concepts

- **Projection** — mapping 3D objects onto a 2D display surface
- **Hidden Surface Removal** — hiding surfaces obscured from the viewer
- **Illumination & Shading** — simulating light reflection for realism
  - *Local illumination* — direct light only
  - *Global illumination* — includes reflected light and transparency

Allied Fields

- **Image Processing** — pixel-based operations on existing images (not synthesizing from scratch)
- **Computer-Human Interaction** — interfaces and logical input devices

Chapter 2: Image Representation

Digital Image Basics

Term	Definition
Resolution	Number of pixels per unit length (e.g., pixels per inch)
Aspect Ratio	Width ÷ Height (e.g., 1920 × 1080 → 16:9)

Color Models

RGB (Additive — for displays)

$$\text{Color} = (R, G, B), \quad R, G, B \in [0, 1]$$

- Black = (0,0,0), White = (1,1,1)
- The **gray axis** is the diagonal from black to white

CMY (Subtractive — for printers)

$$C = 1-R, \quad M = 1-G, \quad Y = 1-B$$

Image Representation Methods

Direct Coding

Each pixel stores actual color values.

Type	Bits/pixel	Colors
Binary (B&W)	1	2
Grayscale	8	256
True color (RGB)	24	~16.7 million

Lookup Table (Color Map)

Each pixel stores an **index** into a color table.

- 8-bit pixel → 256-entry table, each entry = 24-bit RGB
- **Memory formula:** `Entries × 3 bytes`
  - Example: 2-byte pixel → 65,536 entries → 65,536 × 3 = **196,608 bytes**

Output Devices

CRT Monitors

- Electron guns strike phosphor-coated screen → emits light
- Requires refresh (~60 Hz) to avoid flicker
- Uses 3 guns (R, G, B) with shadow masks

Printers (CMYK)

- Paper is white; ink **subtracts** light
- Black ink used separately (cheaper, better quality than mixing C+M+Y)

Techniques to simulate shades:

Technique	Method	Resolution
Halftoning	Varies dot size in a grid	Reduced

Technique	Method	Resolution
Dithering	Threshold matrix decides pixel ON/OFF	Preserved
Error Diffusion	Spreads rounding error to neighbors	Preserved

Dithering rule: Pixel ON if `ImageIntensity(x,y) > D_n(i,j)`, where `i = x mod n`, `j = y mod n`

Error Diffusion (Floyd-Steinberg): Distributes quantization error to unprocessed neighbors → smooth gradients without banding.

Image Files

- Structure: **Header** (format, dimensions) + **Image Data**
- Compression: **Run-Length Encoding (RLE)** — e.g., `AAAAAABBBCC` → `(A,6),(B,3),(C,2)`

Exam Math

Q1a: What is quantization?

Quantization is the process of mapping a continuous range of values into a smaller, finite, discrete set. In image representation, it means converting continuous light intensities into discrete digital values — for example, mapping analog color intensity to an 8-bit value in the range 0–255.

Q1b: Maximum memory for 10 images using direct coding (640×520 pixels, 3 bytes/pixel)?

Given:

- Image dimensions: 640 × 520 pixels
- Color depth: 3 bytes per pixel (1 byte each for R, G, B)
- Number of images: 10

Step 1 — Total pixels per image:
$$640 \times 520 = 332{,}800 \text{ pixels}$$

Step 2 — Memory per image:
$$332{,}800 \text{ pixels} \times 3 \text{ bytes/pixel} = 998{,}400 \text{ bytes}$$

Step 3 — Total memory for 10 images:
$$998{,}400 \times 10 = \boxed{9{,}984{,}000 \text{ bytes}}$$

Converting: $9{,}984{,}000 \div 1024 = 9{,}750 \text{ KB} \approx 9.52 \text{ MB}$

Q1c: How many colors can be stored in a lookup table if each image's index memory must not exceed 500 KB?

Given:

- Image size: 640 × 520 = 332,800 pixels (from Q1b)
- Memory limit per image: 500 KB
- Goal: find maximum number of bits per pixel for the index, then find the number of colors

Step 1 — Convert 500 KB to bytes:
$$500 \times 1024 = 512{,}000 \text{ bytes}$$

Step 2 — Convert bytes to bits:
$$512{,}000 \times 8 = 4{,}096{,}000 \text{ bits}$$

Step 3 — Calculate available bits per pixel:
$$\frac{4{,}096{,}000 \text{ bits}}{332{,}800 \text{ pixels}} \approx 12.307 \text{ bits/pixel}$$

Step 4 — Round down to a whole number (must be integer bit depth):
$$\lfloor 12.307 \rfloor = 12 \text{ bits/pixel}$$

(We round down, not up, to stay within the 500 KB limit.)

Step 5 — Calculate number of representable colors:
$$2^{12} = \boxed{4{,}096 \text{ colors}}$$

Bonus Q: If 2-byte pixel values are used in a 24-bit lookup table, how many bytes does the lookup table occupy?

Given:

- Pixel size: 2 bytes = 16 bits → used as an index
- Each LUT entry: 24-bit color = 3 bytes

Step 1 — Number of LUT entries (one per possible index value):
$$2^{16} = 65{,}536 \text{ entries}$$

Step 2 — Size of each entry:
$$24 \text{ bits} \div 8 = 3 \text{ bytes per entry}$$

Step 3 — Total LUT size:
$$65{,}536 \times 3 = \boxed{196{,}608 \text{ bytes}}$$

Chapter 3: Scan Conversion

Scan-Converting a Point

$$X_{pixel} = \lfloor x + 0.5 \rfloor, \quad Y_{pixel} = \lfloor y + 0.5 \rfloor$$

Example: Map continuous point $(10.3,\ 20.7)$ to a pixel.

Step 1 — Add 0.5 to each coordinate:
$$x + 0.5 = 10.3 + 0.5 = 10.8$$
$$y + 0.5 = 20.7 + 0.5 = 21.2$$

Step 2 — Apply floor:
$$X_{pixel} = \lfloor 10.8 \rfloor = 10, \quad Y_{pixel} = \lfloor 21.2 \rfloor = 21$$

Result: Pixel plotted at $(10,\ 21)$

Line Drawing Algorithms

1. Slope-Intercept Method (y = mx + b)

Steps:

1. Compute slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$
2. Compute y-intercept: $b = y_1 - m \cdot x_1$
3. For each integer x from x_1 to x_2: compute $y = mx + b$, round to nearest integer, plot $(x, \text{round}(y))$

Example — Line from (2, 3) to (6, 5):

Step 1 — Calculate slope:
$$m = \frac{5 - 3}{6 - 2} = \frac{2}{4} = 0.5$$

Step 2 — Calculate y-intercept:
$$b = y_1 - m \cdot x_1 = 3 - (0.5 \times 2) = 3 - 1 = 2$$

Step 3 — Iterate and compute y for each x (using $y = 0.5x + 2$):

x	Calculation	y (float)	$\text{round}(y)$	Plot
2	$0.5(2) + 2 = 1 + 2$	3.0	3	(2, 3)
3	$0.5(3) + 2 = 1.5 + 2$	3.5	4	(3, 4)
4	$0.5(4) + 2 = 2 + 2$	4.0	4	(4, 4)
5	$0.5(5) + 2 = 2.5 + 2$	4.5	5	(5, 5)
6	$0.5(6) + 2 = 3 + 2$	5.0	5	(6, 5)

✅ Advantage	❌ Disadvantage
Simple to understand	Floating-point multiplication every pixel
Direct calculation	Visual gaps when $m > 1$

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2. DDA (Digital Differential Analyzer)

Steps:

1. Compute differences: $dx = x_2 - x_1$, $dy = y_2 - y_1$
2. Determine step count: $\text{step} = \max(|dx|, |dy|)$ — ensures no gaps regardless of slope
3. Compute per-step increments: $x_{inc} = dx / \text{step}$, $y_{inc} = dy / \text{step}$
4. Initialize: $x = x_1$, $y = y_1$
5. For each iteration: plot $(\text{round}(x), \text{round}(y))$, then $x = x + x_{inc}$, $y = y + y_{inc}$

Example — Line from (2, 3) to (6, 5):

Step 1 — Compute differences:
$$dx = 6 - 2 = 4, \quad dy = 5 - 3 = 2$$

Step 2 — Determine step count:
$$\text{step} = \max(|4|, |2|) = \max(4, 2) = 4$$

Step 3 — Compute increments:
$$x_{inc} = \frac{4}{4} = 1.0, \quad y_{inc} = \frac{2}{4} = 0.5$$

Step 4 — Initialize: $x = 2.0,\ y = 3.0$

Step 5 — Iterate (at each step: plot → add increments):

Iter	x	y	$\text{round}(x)$	$\text{round}(y)$	Plot	x_{next}	y_{next}
0	2.0	3.0	$2.0+1.0=3.0$	$3.0+0.5=3.5$	2	3	**(2,3)**
1	3.0	3.5	$3.0+1.0=4.0$	$3.5+0.5=4.0$	3	4	**(3,4)**
2	4.0	4.0	$4.0+1.0=5.0$	$4.0+0.5=4.5$	4	4	**(4,4)**
3	5.0	4.5	$5.0+1.0=6.0$	$4.5+0.5=5.0$	5	5	**(5,5)**
4	6.0	5.0	—	—	6	5	**(6,5)**

✅ Advantage	❌ Disadvantage
No multiplication	Still uses floating-point
No gaps in line	Cumulative rounding error on long lines

3. Bresenham's Line Algorithm (Integer Only)

Steps (for $m < 1$):

1. Compute: $dx = x_2 - x_1$, $dy = y_2 - y_1$
2. Pre-compute constants: $2dy$ and $2dy - 2dx$
3. Initialize: $x = x_1$, $y = y_1$, $P_0 = 2dy - dx$
4. At each step: plot (x, y), increment $x = x + 1$, then:
    - If $P_n < 0$: y stays, $P_{n+1} = P_n + 2dy$

    - If $P_n \geq 0$: $y = y + 1$, $P_{n+1} = P_n + 2dy - 2dx$
5. Repeat until $x = x_2$

Example — Line from (2, 3) to (6, 5):

Step 1 — Compute differences:
$$dx = 6 - 2 = 4, \quad dy = 5 - 3 = 2$$

Step 2 — Pre-compute constants:
$$2dy = 2 \times 2 = 4$$
$$2dy - 2dx = 4 - 2(4) = 4 - 8 = -4$$

Step 3 — Initial decision variable:
$$P_0 = 2dy - dx = 4 - 4 = 0$$

Step 4 — Initialize: plot $(2, 3)$, then iterate:

Step	x	y	x_{next}	Action	y_{next}	p	p_{next}
0	2	3	3	y++	4	0	-4
1	3	4	4	y stays	4	-4	0
2	4	4	5	y++	5	0	-4
3	5	5	6	y stays	5	-4	—

Plotted pixels: (2,3), (3,4), (4,4), (5,5), (6,5)

Final pixels plotted: $(2,3),\ (3,4),\ (4,4),\ (5,5),\ (6,5)$

For $m \ge 1$ (steep lines):

- The roles of x and y swap compared to $m < 1$.
- Increment y each step, and decide whether to increment x based on P.
- Initial $P_0 = 2dx - dy$.
- If $P < 0$: x stays, $P = P + 2dx$.
- If $P \ge 0$: x increments, $P = P + 2dx - 2dy$.
  Memory tip: When slope flips, dx and dy swap in the formulas.

✅ Advantage	❌ Disadvantage
Integer-only (fastest)	More complex to implement for all slopes
No rounding needed	Different cases needed for $m>1$, negatives

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Circle & Ellipse Drawing

8-Way Symmetry (Circles)

Compute one octant ($x \leq y$), mirror to get 8 points:

Octant	Transform	Pixel (from point (2,5))
1	(x, y)	(2, 5)
2	(y, x)	(5, 2)
3	$(-y, x)$	(−5, 2)
4	$(-x, y)$	(−2, 5)
5	$(-x, -y)$	(−2, −5)
6	$(-y, -x)$	(−5, −2)
7	$(y, -x)$	(5, −2)
8	$(x, -y)$	(2, −5)

> Ellipses use **4-way symmetry** only — x and y cannot be swapped.

Performance gain: 87.5% fewer calculations for circles; 75% for ellipses.

Bresenham's Circle Algorithm

Initialize: $x = 0$, $y = r$, $d_0 = 3 - 2r$

At each step (while $x \leq y$):

- Plot 8 symmetric points for current (x, y)
- If $d < 0$: y stays, $d = d + 4x + 6$
- If $d \geq 0$: $y = y - 1$, $d = d + 4x - 4y + 10$
- Always: $x = x + 1$

Example — Circle with $r = 5$, centered at $(0,0)$:

Step 1 — Initialize:
$$x = 0, \quad y = 5$$
$$d_0 = 3 - 2(5) = 3 - 10 = -7$$

Step 2 — Iterate (note: d update uses x and y values *before* incrementing):

Step	x	y	x_{next}	Action	y_{next}	d	d_{next}
Init	0	5	1	d < 0	5	-7	-1
1	1	5	2	d < 0	5	-1	9
2	2	5	3	d >= 0	4	9	7
3	3	4	4	d >= 0	3	7	13

Stop: next $x = 5 > y = 3$, so first octant is complete.

Points computed (x, y): $(0,5), (1,5), (2,5), (3,4), (4,3)$ — each is mirrored using 8-way symmetry to produce the full circle.

Midpoint Circle Algorithm

Initialize: $x = 0$, $y = r$, $p_0 = 1 - r$

At each step (while $x \leq y$):

- Plot 8 symmetric points for current (x, y)
- If $p < 0$: y stays, $p = p + 2x + 3$
- If $p \geq 0$: $y = y - 1$, $p = p + 2x - 2y + 5$
- Always: $x = x + 1$

Example — Circle with $r = 5$, centered at $(0,0)$:

Step 1 — Initialize:
$$x = 0, \quad y = 5$$
$$p_0 = 1 - r = 1 - 5 = -4$$

Step 2 — Iterate (note: p update uses x and y values *before* incrementing x, but *after* decrementing y when applicable):

Step	x	y	x_{next}	Action	y_{next}	p	d_{next}
Init	0	5	1	p < 0	5	-4	-1
1	1	5	2	p < 0	5	-1	4
2	2	5	3	p >= 0	4	4	3
3	3	4	4	p >= 0	3	3	6

Stop: next $x = 5 > y = 3$, first octant complete.

Points computed: $(0,5),\ (1,5),\ (2,5),\ (3,4),\ (4,3)$ — mirrored using 8-way symmetry.

> Midpoint is more adaptable than Bresenham's Circle — it is the standard method used for ellipses. For ellipses, split the curve into two regions at the point where slope $= -1$.

Scan-Converting Characters (Fonts)

Bitmap (Raster) Fonts

- Store characters as pre-calculated binary grids
- Rendering = copy grid to screen memory (BitBlt)

✅ Advantage	❌ Disadvantage
Extremely fast	Blocky when scaled

✅ Advantage	❌ Disadvantage
Pixel-perfect at native size	Separate file per size needed

Outline (Vector) Fonts (e.g., TrueType, OpenType)

- Store mathematical curves (Bézier) and lines
- Rendering pipeline: **Fetch geometry → Transform → Rasterize boundary → Scanline fill → Hint**

✅ Advantage	❌ Disadvantage
Infinitely scalable	Computationally expensive

| Single file for all sizes | Blurry at small sizes without hinting |

Scan-Converting Arcs, Sectors, Rectangles

Arcs

Three approaches:

1. **Trigonometric:** $x = h + r\cos\theta$, $y = k + r\sin\theta$ — accurate but slow (uses sin/cos)
2. **Polynomial:** $y = \sqrt{r^2 - x^2}$ — uses square roots, no symmetry
3. **Bresenham (avoid for arcs):** Must compute entire circle then check bounds — inefficient, risks infinite loop

Sectors (Pie Slice)

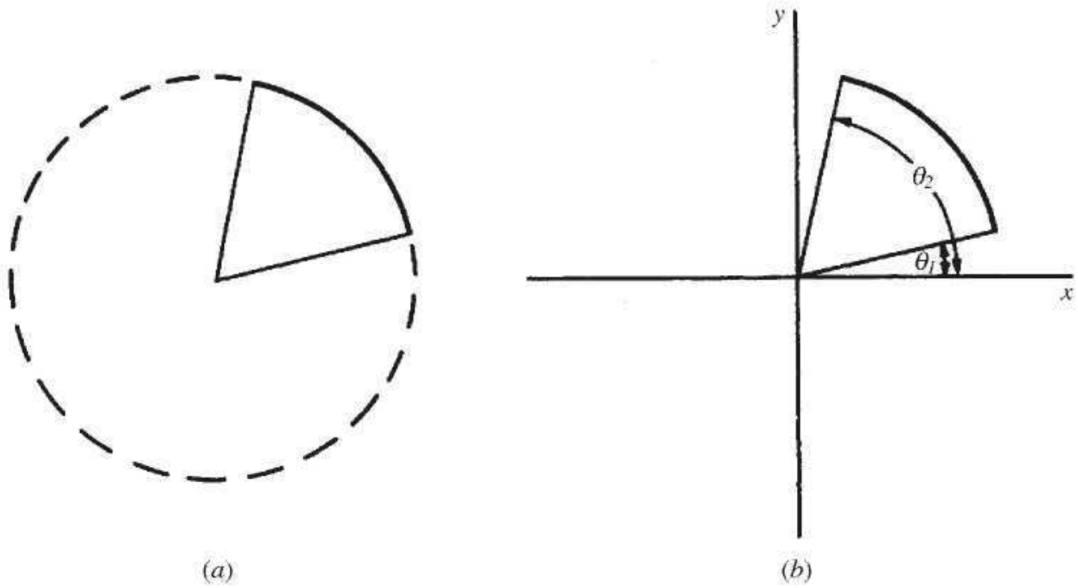

Fig. 3-13

Fig. 3-13: Scan-converting a sector (pie slice): (a) shows the arc and sector, (b) shows the angles and endpoints used for filling.

1. Draw arc from θ_1 to θ_2
2. Calculate endpoints: $(h + r\cos\theta_1,\ k + r\sin\theta_1)$ and $(h + r\cos\theta_2,\ k + r\sin\theta_2)$
3. Draw two lines from center to each endpoint (use Bresenham's line)

Rectangle (Axis-aligned)

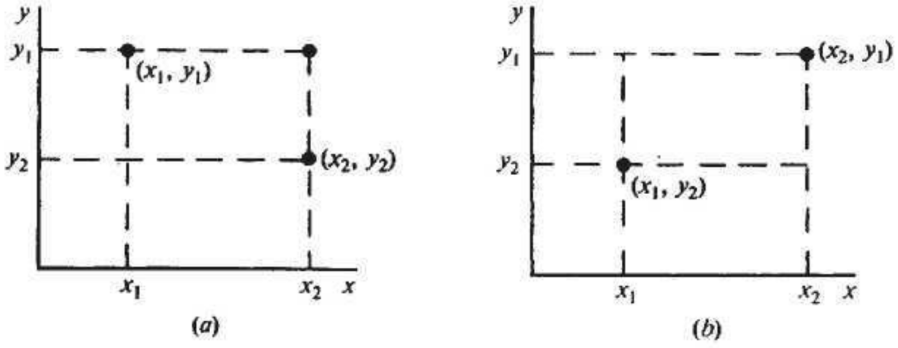

Fig. 3-16

Fig. 3-16: Scan-converting a rectangle: (a) and (b) show how the four corners are used to define the rectangle for filling or drawing edges.

- Input: two opposite corners (x_1, y_1) and (x_2, y_2)
- Derive: (x_1, y_2) and (x_2, y_1)
- Draw 4 edges using any line algorithm

Region Filling

Interior-defined region: A region where filling begins from a seed point inside the boundary and continues outward until the boundary is reached.

Boundary-defined region: A region enclosed by a specific boundary color, where filling begins from a seed point inside the area and continues outward pixel-by-pixel until the specific boundary color is reached.

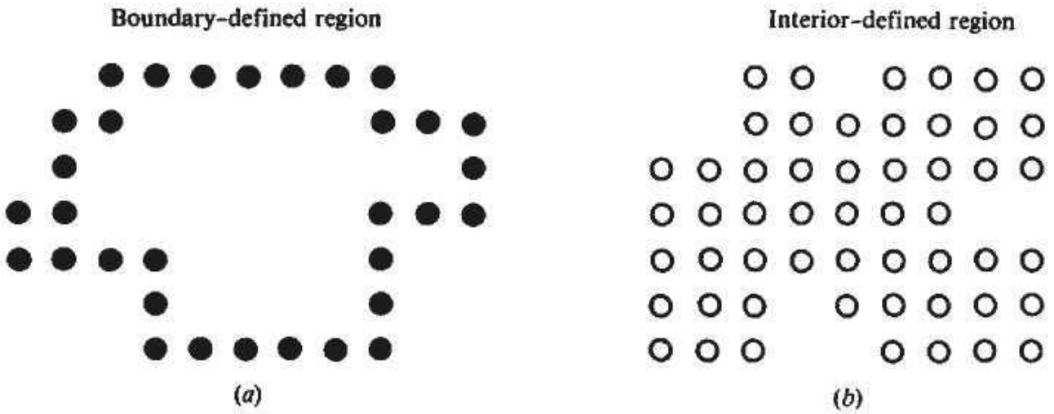

Fig. 3-17

Fig. 3-17: (a) Boundary-defined region: filling is constrained by a boundary color. (b) Interior-defined region: filling spreads outward from a seed point until the boundary is reached.

Pixel Connectivity

Type	Neighbors
4-Connected	Up, Down, Left, Right only
8-Connected	All 8 surrounding pixels (includes diagonals)

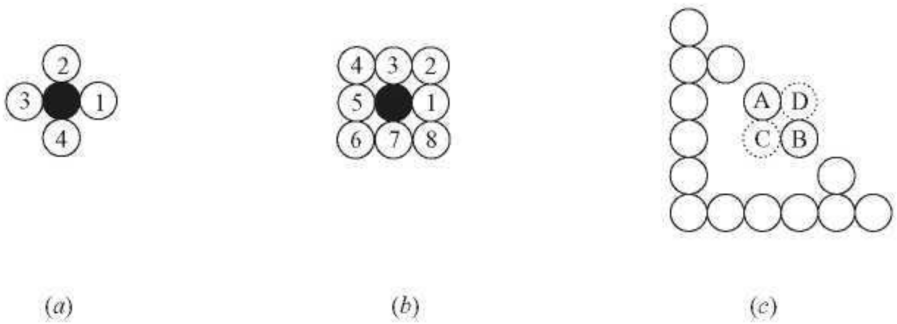

Fig. 3-18 4-connected vs. 8-connected pixels.

Fig. 3-18: (a) 4-connected neighbors: only up, down, left, right. (b) 8-connected neighbors: includes diagonals. (c) Example showing how connectivity affects region filling.

> **Critical rule:** If boundary is 8-connected → fill must be 4-connected, and vice versa (prevents leaking).

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1. Boundary-Fill (Recursive)

Used when region is enclosed by a specific **boundary color**.

Steps:

1. Start at seed pixel
2. If pixel = fill color OR boundary color → stop
3. Else: paint pixel with fill color, recurse on neighbors

✅ Advantage	❌ Disadvantage
Works on irregular shapes	Stack overflow on large regions
Easy to implement	Fails if boundary is multi-colored

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2. Flood-Fill (Recursive)

Used when region is defined by a uniform **interior color** (e.g., paint bucket tool).

Steps:

1. Start at seed, record `Old_Color`
2. If pixel ≠ `Old_Color` OR already = `New_Color` → stop
3. Else: paint with `New_Color`, recurse on neighbors

✅ Advantage	❌ Disadvantage
No boundary color needed	Stack overflow risk
Natural stopping condition	Cannot fill gradients

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3. Scanline Fill (Optimized Polygon Fill)

We represent a polygonal region in terms of a sequence of vertices $V_1, V_2, V_3, \ldots$, that are connected by edges $E_1, E_2, E_3, \ldots$, (see Fig. 3-19). We assume that each vertex V_i has already been scan-converted to integer coordinates (x_i, y_i).

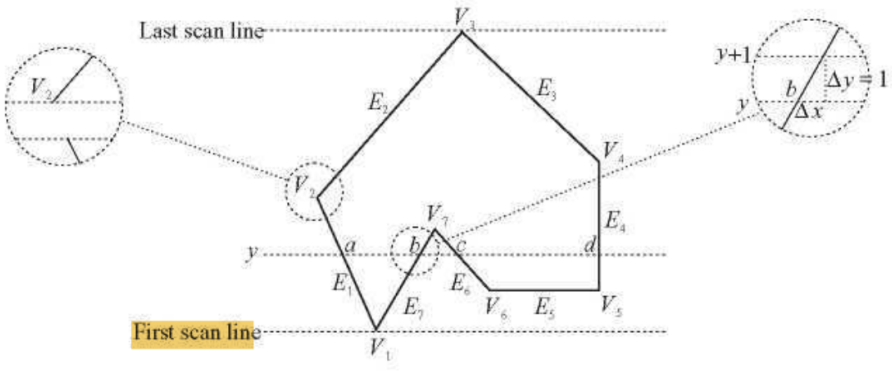


Fig. 3-19 Scan-converting a polygonal region.

Fig. 3-19: Scan-converting a polygonal region.

Table 3-1 An edge list.

Edge	y_{min}	y_{max}	x coordinate of vertex with $y = y_{min}$	$1/m$
E_1	y_1	$y_2 - 1$	x_1	$1/m_1$
E_7	y_1	y_7	x_1	$1/m_7$
E_4	y_5	$y_4 - 1$	x_5	$1/m_4$
E_6	y_6	y_7	x_6	$1/m_6$
E_2	y_2	y_3	x_2	$1/m_2$
E_3	y_4	y_3	x_4	$1/m_3$

Table 3-1: An edge list for scanline fill.

Fills row-by-row using the polygon's geometric edge data. Eliminates recursion.

Core Concepts:

Concept	Explanation
Sweeping Scanline	Process one horizontal line at a time, bottom to top
Vertex Problem	Fix: shorten lower edge by 1 ($y_{max} - 1$) at monotonic vertices
Horizontal Rule	Ignore horizontal edges
Edge Activation Rule	Edge is active from y_{min} up to (but not including) y_{max}; it is deactivated/removed after the scanline at y_{max} is processed.
Odd-Even Rule	Fill pixels between paired intersection points
Edge Coherence	$x_{new} = x_{old} + \frac{1}{m}$ — no recalculation per row

Algorithm Steps:

1. **Build Edge Table (ET):** For each non-horizontal edge compute $y_{min}, y_{max}, x, \frac{1}{m}$; apply $y_{max}-1$ rule; sort by y_{min}
2. **Initialize:** $y = \min(y_{min})$
3. **Update AEL:** Move edges with $y_{min} = y$ into Active Edge List
4. **Sort AEL** by x (left to right)
5. **Fill:** Pair edges (0&1, 2&3, …); fill between each pair
6. **Remove** edges where $y_{max} = y$
7. **Increment:** $y = y + 1$; update $x_{new} = x + \frac{1}{m}$ for remaining AEL edges

8. **Repeat** until ET and AEL are empty

Worked Example — Q2(b): Vertices $(-4,0),(0,2),(4,0),(2,4),(6,4),(0,8),(-6,4),(-4,4)$

Step 1 — Form edges and discard horizontal ones:

Edge	From → To	Horizontal?
E1	$(-4,0)\to(0,2)$	No — keep
E2	$(0,2)\to(4,0)$	No — keep
E3	$(4,0)\to(2,4)$	No — keep
E4	$(2,4)\to(6,4)$	**Yes — discard**
E5	$(6,4)\to(0,8)$	No — keep
E6	$(0,8)\to(-6,4)$	No — keep
E7	$(-6,4)\to(-4,4)$	**Yes — discard**
E8	$(-4,4)\to(-4,0)$	No — keep

Step 2 — Compute ET values using $\frac{1}{m} = \frac{x_2 - x_1}{y_2 - y_1}$:

- **E1** $(-4,0)\to(0,2)$: $y_{min}=0,\ y_{max}=2,\ x=-4,\ \frac{1}{m}=\frac{0-(-4)}{2-0}=\frac{4}{2}=2$
- **E2** $(0,2)\to(4,0)$: $y_{min}=0,\ y_{max}=2,\ x=4,\ \frac{1}{m}=\frac{4-0}{0-2}=\frac{4}{-2}=-2$
- **E3** $(4,0)\to(2,4)$: $y_{min}=0,\ y_{max}=4,\ x=4,\ \frac{1}{m}=\frac{2-4}{4-0}=\frac{-2}{4}=-0.5$
- **E5** $(6,4)\to(0,8)$: $y_{min}=4,\ y_{max}=8,\ x=6,\ \frac{1}{m}=\frac{0-6}{8-4}=\frac{-6}{4}=-1.5$
- **E6** $(0,8)\to(-6,4)$: $y_{min}=4,\ y_{max}=8,\ x=-6,\ \frac{1}{m}=\frac{-6-0}{4-8}=\frac{-6}{-4}=1.5$
- **E8** $(-4,4)\to(-4,0)$: $y_{min}=0,\ y_{max}=4,\ x=-4,\ \frac{1}{m}=\frac{-4-(-4)}{0-4}=\frac{0}{-4}=0$

Step 3 — Final Edge Table (sorted by y_{min}):

Edge	y_{min}	y_{max}	x at y_{min}	$1/m$
E1	0	2	−4	2
E2	0	2	4	−2
E3	0	4	4	−0.5
E8	0	4	−4	0
E5	4	8	6	−1.5
E6	4	8	−6	1.5

Step 4 — Initialize: $y = 0$ (smallest y_{min})

Scanline $y = 0$:

- AEL: add E1, E2, E3, E8 (all have $y_{min} = 0$)
- Sort AEL by x: $-4\ (\text{E8}),\ -4\ (\text{E1}),\ 4\ (\text{E3}),\ 4\ (\text{E2})$
- Pair (E8,E1) → fill from -4 to -4 → zero width, **no fill**
- Pair (E3,E2) → fill from 4 to 4 → zero width, **no fill**
- No edges have $y_{max} = 0$, so none are removed
- Update x values: $x_{new} = x_{old} + \frac{1}{m}$

Edge	Old x	$1/m$	New x
E1	–4	2	$-4 + 2 = -2$
E2	4	–2	$4 + (-2) = 2$
E3	4	–0.5	$4 + (-0.5) = 3.5$
E8	–4	0	$-4 + 0 = -4$

Scanline $y = 1$:

- No new edges enter AEL
- AEL sorted by updated x: $-4\ (\text{E8}),\ -2\ (\text{E1}),\ 2\ (\text{E2}),\ 3.5\ (\text{E3})$
- Pair (E8, E1) → **fill from $x = -4$ to $x = -2$ ✔**
- Pair (E2, E3) → **fill from $x = 2$ to $x = 3.5$ ✔**
- Update x values:

Edge	Old x	$1/m$	New x
E1	–2	2	$-2 + 2 = 0$
E2	2	–2	$2 + (-2) = 0$
E3	3.5	–0.5	$3.5 - 0.5 = 3$
E8	–4	0	$-4 + 0 = -4$

Scanline $y = 2$:

- Remove E1 and E2 ($y_{max} = 2 = y$)
- AEL now: E3, E8 (sorted: $-4,\ 3$)
- **Fill from $x = -4$ to $x = 3$ ✔**
- Update:

Edge	Old x	$1/m$	New x
E3	3	–0.5	$3 - 0.5 = 2.5$
E8	–4	0	-4

Scanline $y = 3$:

- AEL: E3, E8 (sorted: $-4,\ 2.5$)

- **Fill from $x = -4$ to $x = 2.5$ ✔**
- Update: E3 → $2.5 - 0.5 = 2$; E8 → -4

Scanline $y = 4$:

- Remove E3 and E8 ($y_{max} = 4 = y$)
- Add E5 and E6 from ET ($y_{min} = 4$)
- AEL: E5 $(x=6)$, E6 $(x=-6)$ → sorted: $-6,\ 6$
- **Fill from $x = -6$ to $x = 6$ ✔**
- Update: E5 → $6 - 1.5 = 4.5$; E6 → $-6 + 1.5 = -4.5$

(Continue similarly for $y = 5, 6, 7$ until $y = 8$ where E5 and E6 are removed and both ET and AEL are empty.)

✅ Advantage	❌ Disadvantage
No recursion	ET and AEL maintenance overhead
Fast via edge coherence	AEL re-sort needed when edges cross
Handles concave polygons & holes	

Common Mistakes:

- Including horizontal edges in ET
- Forgetting to sort AEL by x each scanline
- Calculating $\frac{1}{m}$ as $\frac{\Delta y}{\Delta x}$ instead of $\frac{\Delta x}{\Delta y}$
- Forgetting $y_{max} - 1$ rule at monotonic vertices

Anti-Aliasing

Aliasing Artifacts

- Visual distortions caused by approximating continuous objects with discrete pixels.

Artifact	Description
Staircase (Jaggies)	Jagged edges on diagonal lines/curves
Unequal Brightness	Diagonals appear dimmer than horizontal/vertical lines. Because distance between two diagonal pixel is larger
Picket Fence Problem	When an object does not fit into, the pixel grid properly
Edge Flickering	Edge vibrate during animation
Moire Pattern	Interferance occur when high frequency texture are sampled

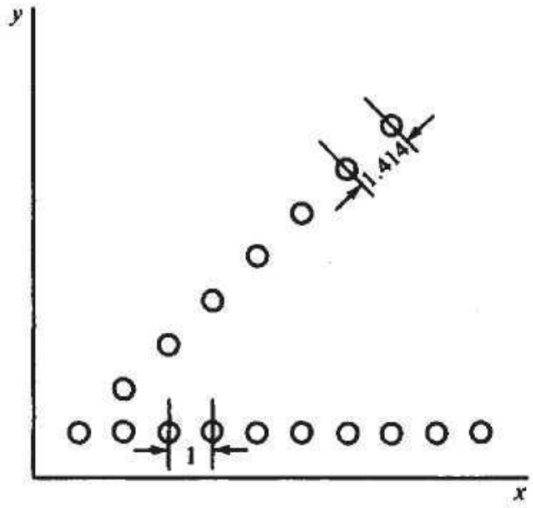


Fig. 3-23

Fig.: Unequal Brightness

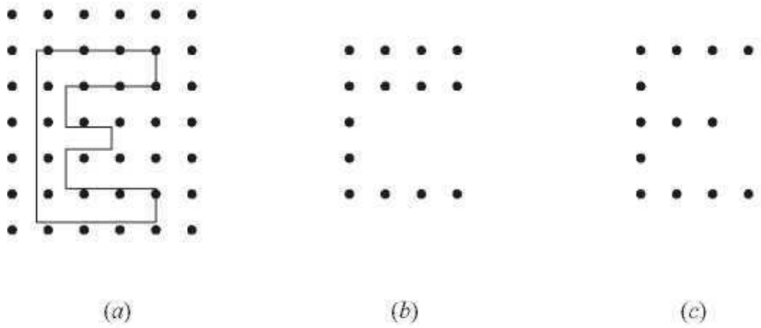


Fig. 3-25 Scan-converting an outline font.

Fig.: Picket Fence Problem with the outline font.
Suppose we want to scan-convert the uppercase character "E" in (a) from its outline description to a bitmap consisting of pixels inside the region defined by the outline. The result in (b) exhibits both asymmetry (the upper arm of the character is twice as thick as the other parts) and dropout (the middle arm is absent). A slight adjustment and/or realignment of the outline can lead to a reasonable outcome (c).

. Why Anti-Aliasing is Needed

- Simply increasing resolution reduces jaggedness but **requires a lot of memory and processing**.
- Anti-aliasing **smooths edges without increasing resolution**, making images appear natural.

. Main Anti-Aliasing Techniques with Examples

A. Pre-Filtering

- Works **before converting to pixels**.
- Each pixel's color is set **based on how much of it is covered by the object**.

Example:

- Pixel partially covered by a white line:

  Coverage = 50% (0.5)
  Object color = White (1)
  Background color = Black (0)
  Pixel color = 1 × 0.5 + 0 × 0.5 = 0.5 → gray

- Fully inside pixel → full white, outside → black.

Result: Smooth edge; shape remains sharp in the center.

B. Supersampling (Post-Filtering)

- Each pixel is divided into **subpixels** (e.g., 3×3).
- Count subpixels inside the object → average their colors → final pixel color.

Example:

- Pixel divided into 3×3 = 9 subpixels
- 6 subpixels are inside a line → coverage = 6/9 ≈ 0.67
- Object color = white (1), background = black (0)
- Pixel color = 1 × 0.67 + 0 × 0.33 = 0.67 → gray

Visual Idea:

[• • •]
[• o o]
[o o o]

- • = subpixel inside object
- o = subpixel outside object
- Coverage = 6/9 → pixel color 0.67 (blended)

Result: Smooth diagonal lines and edges.

C. Pixel Phasing

- Hardware-based: shifts pixels **slightly from their normal positions** to align with true line/contour.
- Reduces stair-step effect without blurring edges.

Example:

- A diagonal line across a pixel grid may look jagged.

- Pixel positions are shifted **fractionally toward the line**, so the line looks smoother.

Visual Idea (simplified):

Before: After:
• o • •
o • o •

- Line edges appear aligned → stair-step effect minimized.

D. Gray-Level / Color Blending

- Uses **intermediate shades or colors** between object and background.
- Only edge pixels are blended, making transitions smooth.

Example:

- White line on black background
- Edge pixel = 50% covered → gray
- Edge pixel = 25% covered → dark gray

Visual Idea:

```
Background → Dark Gray → Gray → White (center of line)
```

Result: Human eye sees a smooth, continuous line.

In short:
Anti-aliasing tricks your eyes by **blending colors, averaging sub-pixels, or shifting pixels slightly**, so edges appear **smooth and natural**, even on low-resolution screens.

Chapter 4: Two-Dimensional Transformations

Introduction
Transformation is the mathematical process of simulating the spatial manipulation of objects. There are two primary viewpoints for transformations:

1. **Geometric Transformation:** The coordinate system remains stationary, and the object itself is moved or altered.
2. **Coordinate Transformation:** The object remains stationary, and the coordinate system is moved or altered around it.

4.1 Geometric Transformations
This section defines the fundamental operations for manipulating a point $P(x,y)$ to a new position $P'(x',y')$:

- **Translation (T_v):** Displacing an object by a specific distance and direction using a vector.

- $x' = x + t_x$ and $y' = y + t_y$

displacement is given by the vector $\mathbf{v} = t_x\mathbf{I} + t_y\mathbf{J}$, the new object point $P'(x', y')$ can be found by applying the transformation $T_\mathbf{v}$ to $P(x, y)$ (see Fig. 4-2).

$$P' = T_\mathbf{v}(P)$$

where $x' = x + t_x$ and $y' = y + t_y$.

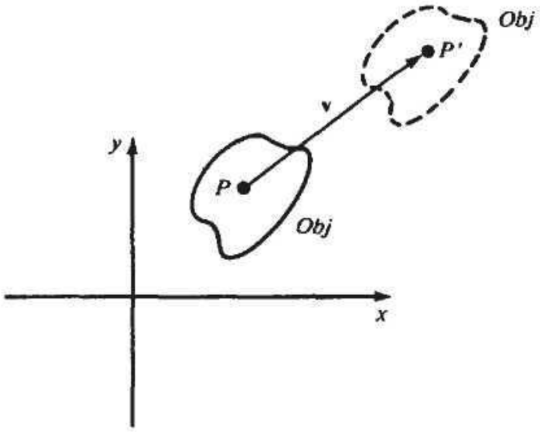


Fig. 4-2

Fig. 4-2: Translation of an object by vector $\vec{v}$, moving point P to P'.

- **Rotation (R_θ):** Rotating an object θ° about the origin. Counterclockwise is positive, and clockwise is negative.
    - $x' = x\cos\theta - y\sin\theta$ and $y' = x\sin\theta + y\cos\theta$

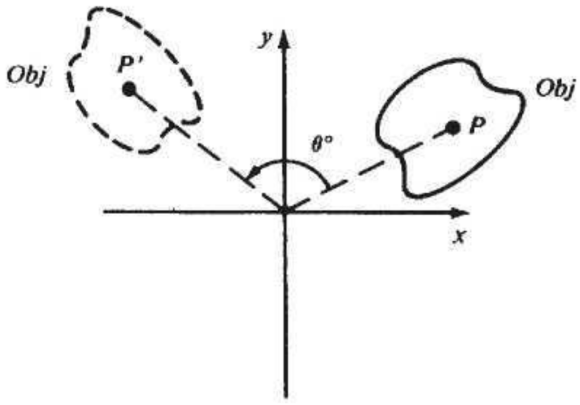


Fig. 4-3

Fig. 4-3: Rotation of an object by θ degrees, moving point P to P'.

- **Scaling (S_{s_x, s_y}):** Expanding or compressing an object using scaling constants s_x and s_y. Only the origin remains fixed. Uniform scaling occurs when $s_x = s_y$.
    - $x' = s_x x$ and $y' = s_y y$

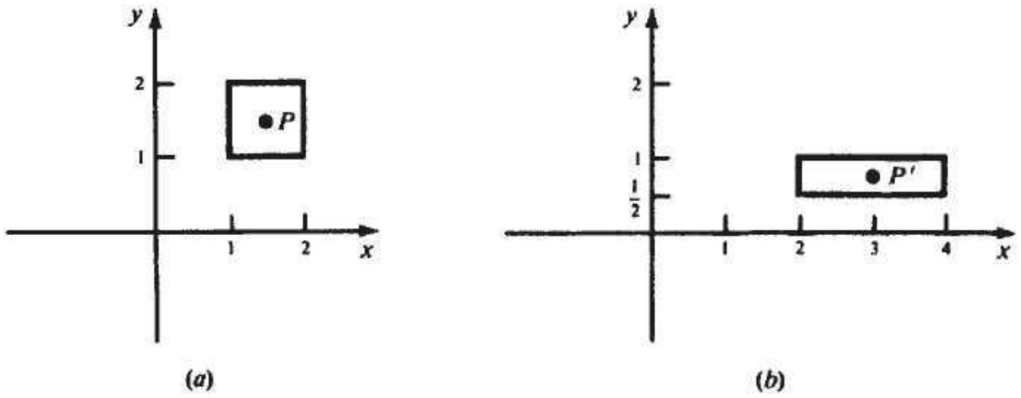


Fig. 4-4

Fig. 4-4: scaling transformation with scaling factors $s_x = 2$ and $s_y = 1/2$.

- **Mirror Reflection (M):** Creating a mirrored image of an object across an axis (e.g., about the x-axis: $x' = x$ and $y' = -y$).

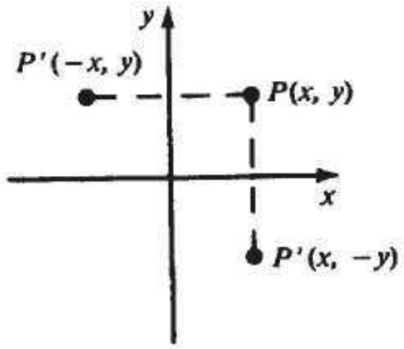


Fig. 4-5

Fig. 4-5: Mirror reflection of a point $P(x, y)$ across the x-axis and y-axis, resulting in $P'(x, -y)$ and $P'(-x, y)$.

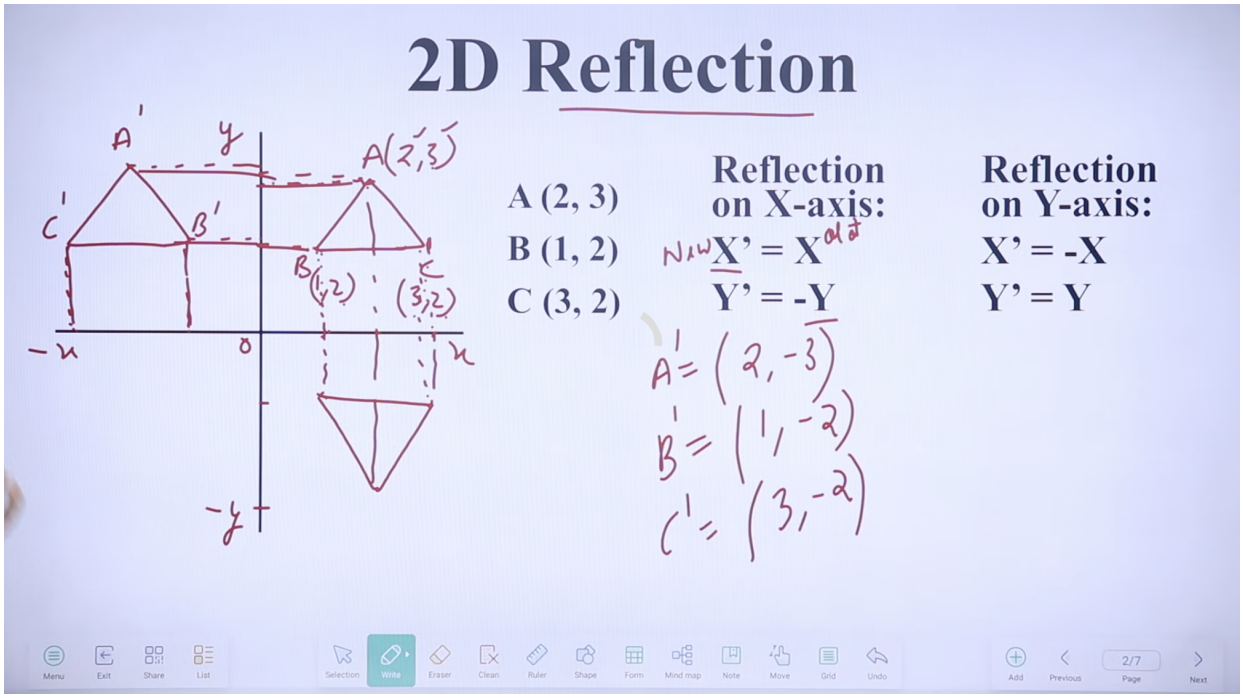


Fig. 4-5b: Example of 2D reflection of a triangle about the x-axis and y-axis, showing coordinate changes and formulas for both axes.

- **Inverse Transformations:** Every geometric transformation has a direct inverse that reverses the operation (e.g., the inverse of translating by v is translating by $-v$).
    - **Translation:** T_{-v} is the inverse of T_v (translation by $-v$ reverses translation by v).
    - **Rotation:** $R_{-\theta}$ is the inverse of R_{θ} (rotation by $-\theta$ undoes rotation by θ).
    - **Scaling:** $S_{1/s_x, 1/s_y}$ is the inverse of S_{s_x, s_y} (scaling by the reciprocal factors undoes the original scaling).
    - **Mirror Reflection:** The inverse of a mirror reflection is itself (applying the same reflection twice returns the object to its original position).

2D Shearing

Shearing is a transformation that slants the shape of an object. It is similar to how normal text becomes italicized. Shearing can be performed along the x-axis or y-axis, and the equations change depending on which axis is being tilted.

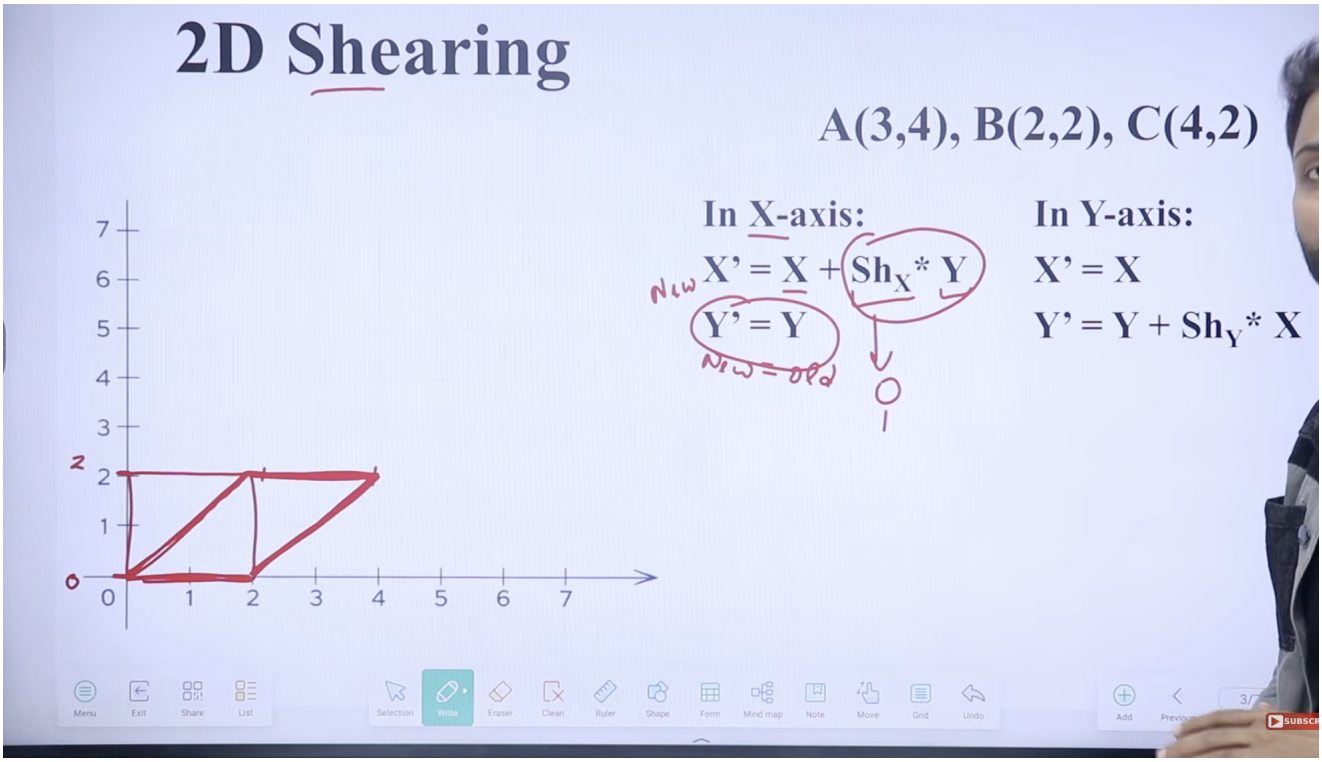


Fig. 4-6: Shearing a triangle along the X-axis. The x coordinates are updated as $x' = x + Sh_x \cdot y$ while y remains unchanged.

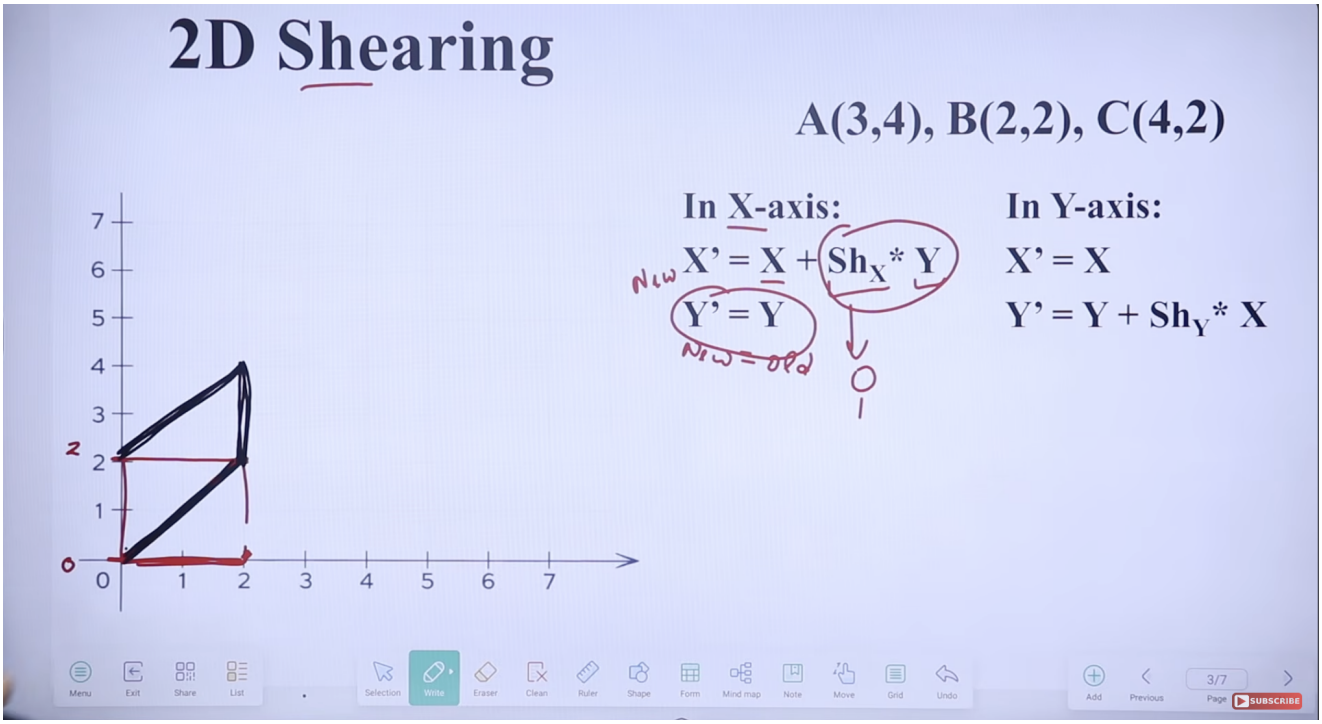


Fig. 4-7: Shearing a triangle along the Y-axis. The y coordinates are updated as $y' = y + Sh_y \cdot x$ while x remains unchanged.

1. Shearing along the X-axis:

- $x' = x + (Sh_x \cdot y)$
- $y' = y$

2. Shearing along the Y-axis:

- $x' = x$
- $y' = y + (Sh_y \cdot x)$

Key Point: The axis not mentioned in the question is the one that remains fixed.

Practical Example:
Given triangle vertices $A(3,4)$, $B(2,2)$, $C(4,2)$:

For X-axis Shearing ($Sh_x = 1$):

- $A':\ x' = 3 + (1 \times 4) = 7 \rightarrow (7, 4)$
- $B':\ x' = 2 + (1 \times 2) = 4 \rightarrow (4, 2)$
- $C':\ x' = 4 + (1 \times 2) = 6 \rightarrow (6, 2)$

For Y-axis Shearing ($Sh_y = 1$):

- $A':\ y' = 4 + (1 \times 3) = 7 \rightarrow (3, 7)$
- $B':\ y' = 2 + (1 \times 2) = 4 \rightarrow (2, 4)$
- $C':\ y' = 2 + (1 \times 4) = 6 \rightarrow (4, 6)$

> The core takeaway: Identify which axis is fixed and apply the corresponding formula.

4.2 Coordinate Transformations

Instead of modifying the object's points, a new coordinate system ($x'y'$) is defined relative to the old one (xy). Transformations (translation, rotation, scaling, and reflection) are applied to the axes, which inversely affects the coordinate descriptions of the stationary points.

4.3 Composite Transformations & Homogeneous Coordinates

- **Composition:** Complex transformations are built by stringing together basic transformations. For example, to magnify an object while keeping its center $C(h,k)$ fixed, you must: (1) Translate the object so C is at the origin, (2) Scale it, and (3) Translate it back to its original position.
- **Matrix Representation:** Rotation, scaling, and reflection can be represented as 2×2 matrices, but translation cannot.
- **Homogeneous Coordinates:** To allow all transformations (including translation) to be treated as matrix multiplications, points are represented as 3D vectors $[x, y, 1]^T$, and transformations are upgraded to **3×3 matrices**.
- **Composite Transformation Matrix (CTM):** Because all basic transformations are now 3×3 matrices, they can be multiplied (concatenated) together into a single CTM, which is highly computationally efficient.

4.4 Instance Transformations

- **Instances:** A complex picture often uses the same object multiple times (e.g., a wheel on a car). An object is defined once in its own local coordinate system, and an **instance transformation** converts those local coordinates into the master "picture" coordinates. This typically involves scaling, rotating, and translating the object into its proper place.

- **Multilevel/Nested Structures:** Objects can be made of sub-objects (e.g., apples on a branch, branch on a tree). To render these efficiently, instance transformations are concatenated across levels so that the lowest-level object is transformed directly into the final picture coordinates in one step.

Here is an exam-oriented summary of Chapter 5, which focuses on selecting and mapping a portion of a 2D scene onto a display device, along with the algorithms required to trim objects that fall outside the viewing area.

Chapter 5: Two-Dimensional Viewing and Clipping

Introduction to Viewing

- **World Coordinate System (WCS):** The master coordinate system where the entire scene is modeled.
- **Window:** A rectangular region in the WCS used to select the specific portion of the scene to be displayed.
- **Normalized Device Coordinate System (NDCS):** A device-independent coordinate system (typically a 1×1 unit square) that represents the virtual display area.
- **Viewport:** A specific rectangular sub-region within the NDCS where the contents of the window are mapped.
- **Viewing Transformation:** The overall process of mapping coordinates from the WCS to the physical device. It involves two steps: **Window-to-Viewport mapping** (WCS to NDCS) and **Workstation transformation** (NDCS to actual device/pixel coordinates).

5.1 Window-to-Viewport Mapping

- This mapping translates and scales coordinates (wx, wy) from the window to coordinates (vx, vy) in the viewport.
- It operates on the principle of maintaining relative placement (e.g., if a point is dead center in the window, it must be dead center in the viewport).
- **Geometric Distortion:** Occurs if the aspect ratio (width-to-height ratio) of the window does not match the aspect ratio of the viewport, causing objects to stretch or compress.

5.2 & 5.3 Point and Line Clipping

Clipping is the computational process of removing the parts of an object that lie outside a specified boundary, known as the clipping window. Only the visible parts inside the window are kept and rendered.

- **Point Clipping:** Simply evaluates if a point's coordinates satisfy $x_{min} \le x \le x_{max}$ and $y_{min} \le y \le y_{max}$.
- **Cohen-Sutherland Line Clipping:**
  - Assigns a **4-bit region code** (Top, Bottom, Right, Left) to each endpoint of a line based on its position relative to the window. (e.g., 0000 means inside).
  - *Trivial Accept:* If both codes are 0000, the line is fully visible.
  - *Trivial Reject:* If the bitwise logical AND of the two codes is not 0000, the line is completely outside.

    - *Candidate:* If neither, the line intersects a boundary. The algorithm iteratively finds the intersection point, clips the outside portion, and re-evaluates the new endpoint's code.
- **Midpoint Subdivision Line Clipping:**
    - Uses a binary search approach. A line spanning the boundary is repeatedly bisected at its midpoint into two segments. Segments are categorized using region codes until they are reduced to pixel-sized visible fragments or discarded as invisible.
- **Liang-Barsky Line Clipping:**
    - A highly efficient algorithm based on the **parametric equation of a line** ($x = x_1 + \Delta x \cdot u$, $y = y_1 + \Delta y \cdot u$, for $0 \le u \le 1$).
    - It calculates the intersection parameter u for all four window boundaries. It tracks the maximum entering u value and the minimum exiting u value to determine the visible segment of the line.

5.4 Polygon Clipping

- **Sutherland-Hodgman Algorithm:**
    - A pipeline approach that clips an entire polygon against one window boundary edge at a time.
    - For each polygon edge, it evaluates 4 cases: (1) Both ends inside $\rightarrow$ output 2nd point. (2) Both outside $\rightarrow$ output nothing. (3) Inside to outside $\rightarrow$ output intersection point. (4) Outside to inside $\rightarrow$ output intersection, then 2nd point.
    - *Drawback:* Can produce unwanted "extraneous edges" connecting disjoint parts when a concave polygon is clipped into multiple pieces.
- **Weiler-Atherton Algorithm:**
    - Solves the extraneous edge problem.
    - It works by tracing the border of the subject polygon. When the border exits the clipping window, the algorithm makes a "right turn" to trace the clipping window's border instead. When it re-enters, it turns back to trace the subject polygon. This creates perfectly closed sub-polygons for output.

5.5 The 2D Graphics Pipeline & Animation Concepts

- **The Pipeline:** Data flows through sequential stages: *Object Definition $|rightarrow$ Modeling Transformation $|rightarrow$ Viewing Transformation $|rightarrow$ Scan Conversion $|rightarrow$ Frame Buffer Display*.
- **Panning & Zooming:** Panning is achieved by translating the window across the WCS. Zooming is achieved by decreasing (zoom in) or increasing (zoom out) the size of the window relative to the viewport.
- **Double Buffering:** An animation technique using two frame buffers. The system displays one buffer while drawing the next frame invisibly in the second buffer. Once drawn, the buffers are swapped to eliminate screen flicker.
- **Lookup Table Animation:** Also known as color cycling. Instead of redrawing an object to move it, the object is pre-drawn in multiple frames. The lookup table is then updated sequentially to change the colors of the frames from "background color" to "visible color", simulating fast, smooth motion.

Introduction

Moving from 2D to 3D graphics requires manipulating objects in three-dimensional space. Just like in 2D, this is achieved through transformations, but it requires upgrading from 3×3 to **4×4 homogeneous coordinate matrices** so that translation, scaling, and rotation can all be combined via matrix multiplication.

6.1 Geometric Transformations

This perspective assumes the coordinate system remains stationary while the object itself is moved.

- **Translation:** Displacing an object by a given distance and direction, prescribed by a 3D vector $V = aI + bJ + cK$. In 4×4 matrix form, the displacement values $a, b,$ and c occupy the rightmost column.
- **Scaling:** Expanding or compressing an object with respect to the origin using three scale factors: s_x, s_y, and s_z. If the scale factor $s > 1$, it is a magnification; if $s < 1$, it is a reduction.
- **Rotation:** 3D rotation is significantly more complex than 2D rotation. While 2D rotation only requires an angle and a center point, 3D rotation requires an angle (θ) and an **axis of rotation**.
    - *Canonical Rotations:* These are rotations around one of the standard positive coordinate axes (the x-axis, y-axis, or z-axis). The direction of a positive angle is determined by the right-hand rule with respect to the axis of rotation.
    - *Arbitrary Axis Rotation:* Rotating an object around a non-standard axis requires a sequence of transformations: translating the object to the origin, rotating the axis so it aligns with a standard axis (like the z-axis), performing the desired rotation, and then applying the inverse operations to place the object back.

6.2 Coordinate Transformations

This perspective assumes the object remains perfectly stationary while the observer (or the coordinate system) is moved around it.

- Calculating the new coordinates of an object relative to the shifted observer is mathematically equivalent to applying the **inverse** geometric transformation to the object. (e.g., If the observer moves 5 units forward, it is mathematically identical to the object moving 5 units backward).

6.3 Composite Transformations

- Complex movements are built by stringing together basic transformations (translation, scaling, and rotation) through a process called **composition of functions**, which in computer graphics is handled via matrix multiplication (concatenation).
- Because 3D transformations use 4×4 matrices, they can all be seamlessly multiplied together to form a single Composite Transformation Matrix (CTM), allowing complex multi-step manipulations to be calculated in a single mathematical step.

6.4 Instance Transformations

- In complex 3D scenes, an object (like a chair) is typically modeled only once in its own local "object coordinate space."

- To place multiple copies (instances) of this chair into a master "scene coordinate space," an **instance transformation** is applied. This is a specific composite transformation matrix that scales, rotates, and translates the generic chair into its specific final position, size, and orientation within the room.

Here is an exam-oriented summary of Chapter 7, which introduces the mathematics and fundamental concepts of projecting 3D objects onto a 2D display surface.

Chapter 7: Mathematics of Projection

1. What is Projection?

Projection is the process of **mapping a 3D point P(x, y, z) onto a 2D image point P′(x′, y′, z′)** on a flat surface called the **projection plane** (or **view plane**).

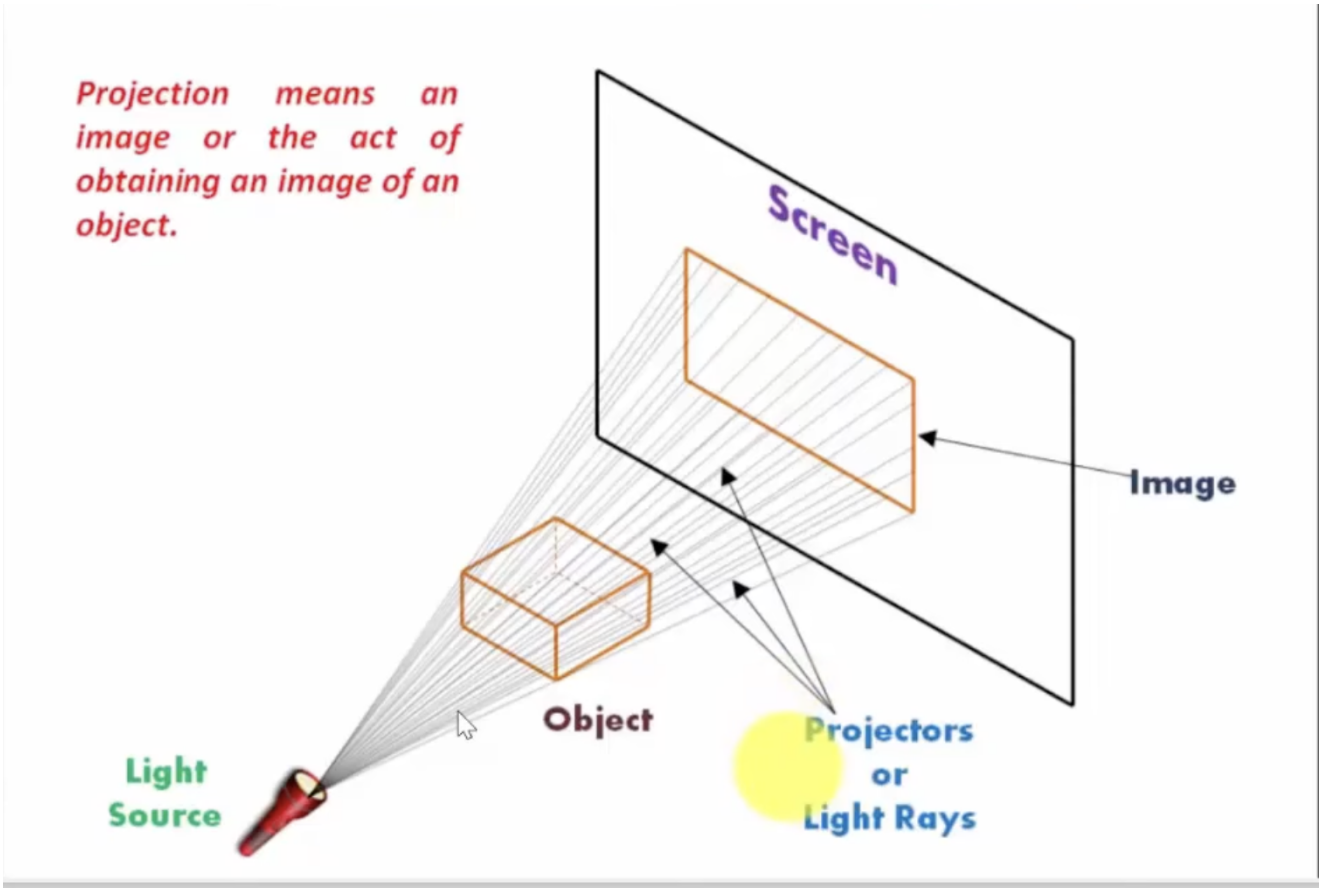
It acts as a bridge between the 3D world and the 2D display on a screen or paper.

The mapping is done through a line called a **projector** — it passes through the 3D object point and intersects the view plane. The intersection point is the projected image P′.

> **Key Observation:** Projection preserves lines. The image of a line segment is itself a line segment (or a point), never a curve.

---

2. Key Elements of Projection



Four components are always needed:

Component	Description
Observer / COP	The viewpoint or Center of Projection where projection rays originate
Object	The 3D entity being projected
Projection Plane (View Plane)	The 2D surface where the final image forms
Projectors	The rays/lines that travel from the object toward the projection plane

3. Taxonomy of Projection

Perspective Projection:

- A method of projecting 3D points onto a plane **so that projection lines converge at a point (the eye/COP)**, making distant objects appear smaller.

Parallel Projection:

- A method of projecting 3D points onto a plane **using parallel lines**, so object sizes remain unchanged regardless of distance.

```
Projections
├── Perspective  (converging projectors — COP at finite distance)
│   ├── One-point perspective    (1 principal vanishing point)
│   ├── Two-point perspective    (2 principal vanishing points)
│   └── Three-point perspective  (3 principal vanishing points)
│
└── Parallel  (parallel projectors — COP at infinity)
    ├── Orthographic  (projectors perpendicular to view plane)
    │   ├── Multiview    (view plane parallel to principal planes)
    │   └── Axonometric  (view plane NOT parallel to principal planes)
    │       ├── Isometric   (equal angles with all 3 axes)
    │       ├── Dimetric    (equal angles with exactly 2 axes)
    │       └── Trimetric   (unequal angles with all 3 axes)
    │
    └── Oblique  (projectors NOT perpendicular to view plane)
        ├── Cavalier  (lines perpendicular to view plane at full length)
        └── Cabinet   (lines perpendicular to view plane at half length)
```

Core difference:

	Perspective	Parallel
COP location	Finite point	At infinity

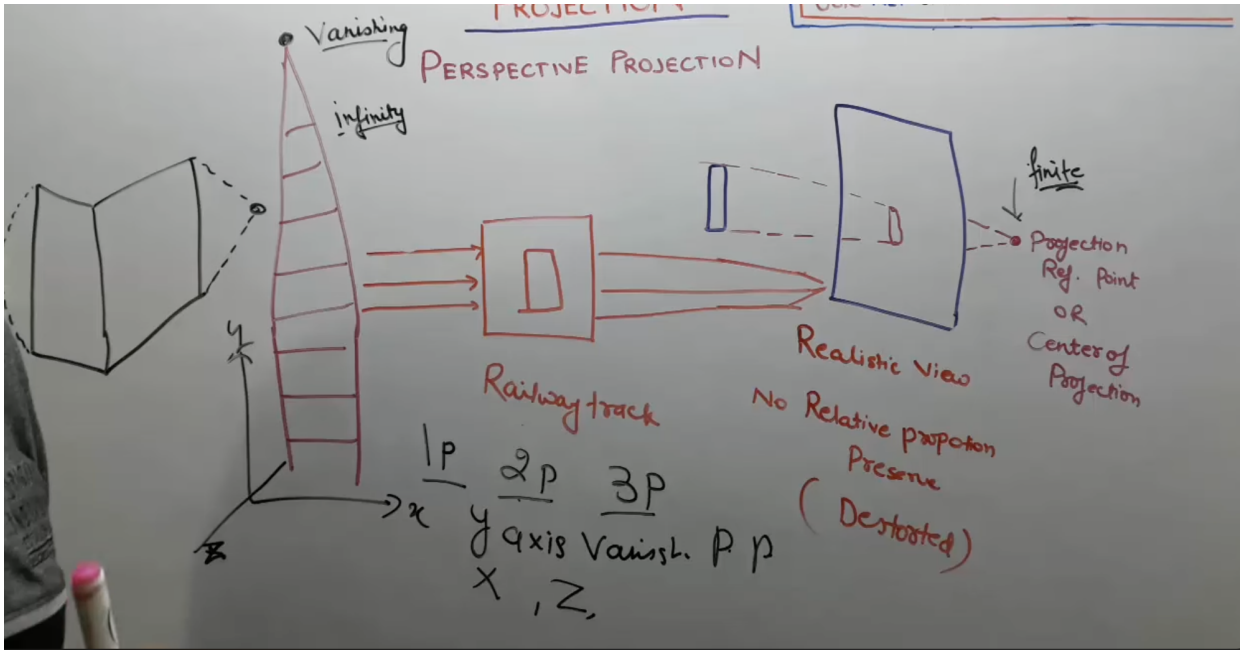
	Perspective	Parallel
Projectors	Converge at COP	All parallel
Realism	High (mimics the eye/camera)	Low
Preserves measurements	No	Yes
Parallel lines stay parallel?	No — converge at vanishing points	Yes
Used for	Artistic/realistic rendering	Engineering/architectural drawings

4. Perspective Projection

In perspective projection, all projectors **converge at a single point** — the **Center of Projection (COP)**, analogous to the eye or camera lens.

A perspective transformation is fully defined by two things:

1. The **COP** — where the rays start from
2. The **View Plane** — defined by a reference point R_0 and a normal vector N



4.1 Reference Point R_0 and Normal Vector N

What is the Reference Point R_0?

R_0 is a known point that lies on the view plane. It acts as the **base location** from which the plane's position in 3D space is defined.

- Think of it as: **the origin of the viewing coordinate system** — the anchor point of the camera/view setup.
- Every projected point P′ is calculated relative to this reference.

- It is used to write the plane equation: any point P′ is on the plane if the vector $(P' - R_0)$ is perpendicular to N.

What is the Normal Vector N?

The normal vector N = (n_1, n_2, n_3) is a vector that is **perpendicular (at 90°) to the plane**.

```
        ^ N  (normal vector — sticks straight out of the plane)
        |
  ------+------
 /             \
/   VIEW PLANE  \
```

Why the normal matters:

- It defines the **orientation (tilt)** of the view plane in 3D space.
- Without N, we cannot tell if the plane faces left, right, up, or diagonally.
- Combined with R_0, it fully describes the plane mathematically.

Plane Equation from R_0 and N:

A point P′(x′, y′, z′) lies on the plane if and only if:

$$N \cdot (P' - R_0) = 0$$

Expanding:

$$n_1(x' - x_0) + n_2(y' - y_0) + n_3(z' - z_0) = 0$$
$$n_1x' + n_2y' + n_3z' = n_1x_0 + n_2y_0 + n_3z_0$$

Define the constant:

$$D = n_1x_0 + n_2y_0 + n_3z_0$$

So the plane equation becomes:

$$n_1x' + n_2y' + n_3z' = D$$

> Left side depends on any candidate point P′. Right side D is a fixed constant for the plane.
> Any point P′ lies on the plane if and only if it satisfies this equation.

4.2 General Perspective — COP at Origin, Arbitrary Plane

Setup:

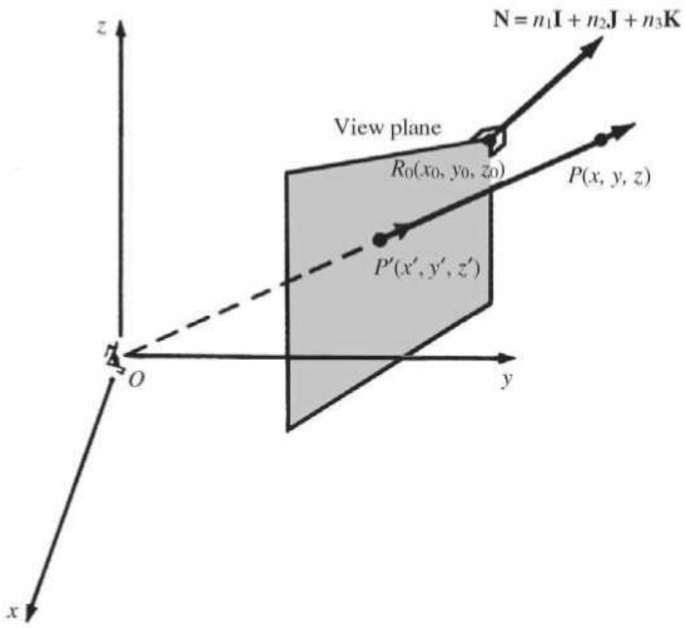
- COP at origin (0, 0, 0)
- View plane defined by normal N = (n_1, n_2, n_3) and reference point $R_o(x_o, y_o, z_o)$
- Object point: P(x, y, z)
- Find: projected point P′(x′, y′, z′)

7.3 Using the origin as the center of projection, derive the perspective transformation onto the plane passing through the point $R_0(x_0, y_0, z_0)$ and having the normal vector $\mathbf{N} = n_1 I + n_2 J + n_3 K$.

SOLUTION

Let $P(x, y, z)$ be projected onto $P'(x', y', z')$. From Fig. 7-16, the vectors $\overline{\mathbf{PO}}$ and $\overline{\mathbf{P'O}}$ have the same direction. Thus there is a number α so that $\overline{\mathbf{P'O}} = \alpha\overline{\mathbf{PO}}$. Comparing components, we have

$$x' = \alpha x \qquad y' = \alpha y \qquad z' = \alpha z$$



Step 1 — Ray from COP through P:

Since the COP is the origin and the ray passes through P, any point on this ray is a scalar multiple of P:

```
P′ = α · P   →   x′ = αx,   y′ = αy,   z′ = αz
```

(α is an unknown scalar to be determined.)

Step 2 — P′ must lie on the plane:

So, it it shoudl satisfy the plane equation $n_1x' + n_2y' + n_3z' = D$

```
n₁(αx) + n₂(αy) + n₃(αz) = D
α(n₁x + n₂y + n₃z) = D
α = D / (n₁x + n₂y + n₃z)
```

Step 3 — Substitute α back to get projection formulas:

$$x' = Dx / (n_1x + n_2y + n_3z)$$
$$y' = Dy / (n_1x + n_2y + n_3z)$$
$$z' = Dz / (n_1x + n_2y + n_3z)$$

where $D = n_1x_0 + n_2y_0 + n_3z_0$.

Step 4 — Matrix form using homogeneous coordinates:

$$\begin{vmatrix} D & 0 & 0 & 0 \\ 0 & D & 0 & 0 \\ 0 & 0 & D & 0 \\ n_1 & n_2 & n_3 & 0 \end{vmatrix} \times \begin{vmatrix} x \\ y \\ z \\ 1 \end{vmatrix} = \begin{vmatrix} Dx \\ Dy \\ Dz \\ n_1x + n_2y + n_3z \end{vmatrix}$$

Divide each of the first three results by the last (perspective divide) to recover x′, y′, z′.

4.3 Standard Perspective — COP at Origin, View Plane z = d

7.4 Find the perspective projection onto the view plane $z = d$ where the center of projection is the origin (0, 0, 0).

SOLUTION

The plane $z = d$ is parallel to the xy plane (and d units away from it). Thus the view plane normal vector $\mathbf{N}$ is the same as the normal vector $\mathbf{K}$ to the xy plane, that is, $\mathbf{N} = \mathbf{K}$. Choosing the view reference point as $R_0(0, 0, d)$, then from Prob. 7.3, we identify the parameters

$$\mathbf{N}(n_1, n_2, n_3) = (0, 0, 1) \qquad R_0(x_0, y_0, z_0) = (0, 0, d)$$

So

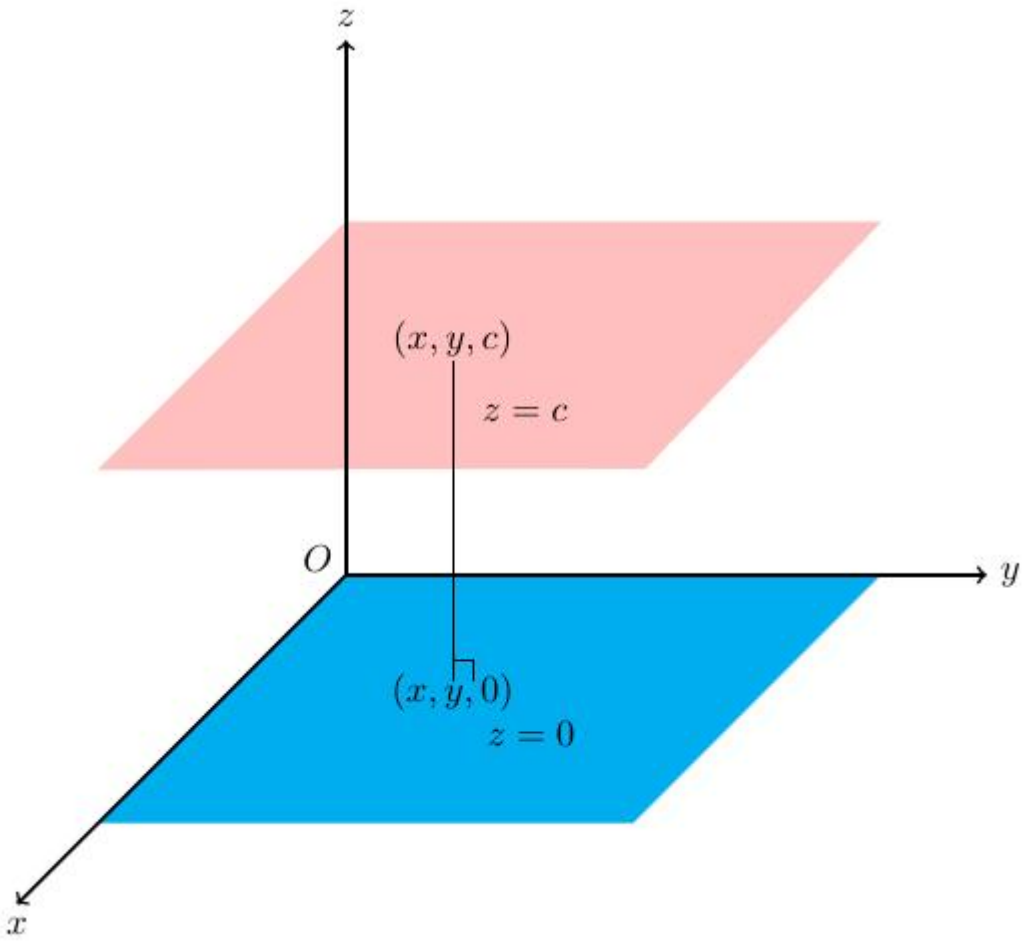
$$d_0 = n_1x_0 + n_2y_0 + n_3z_0 = d$$

and then the projection matrix is

$$Per_{\mathbf{K},R_0} = \begin{pmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

This is a **special case** of Section 4.2 where the plane is z = d.

- Normal: N = (0, 0, 1)
- Reference point: R_0 = (0, 0, d)



Compute D:

```
D = (0)(0) + (0)(0) + (1)(d) = d
```

Substitute into general formulas:

```
x′ = dx / (0·x + 0·y + 1·z) = dx/z
y′ = dy/z
z′ = d
```

Result:

```
x′ = dx/z,   y′ = dy/z,   z′ = d
```

Matrix (homogeneous):

```
| d   0   0   0 |
| 0   d   0   0 |
```

```
| 0   0   0   0 |
| 0   0   1   0 |
```

Apply to (x, y, z, 1) → result is (dx, dy, 0, z). Divide by z → x′ = dx/z, y′ = dy/z.

4.4 Standard Perspective — COP at C(0, 0, −d), View Plane = xy-plane

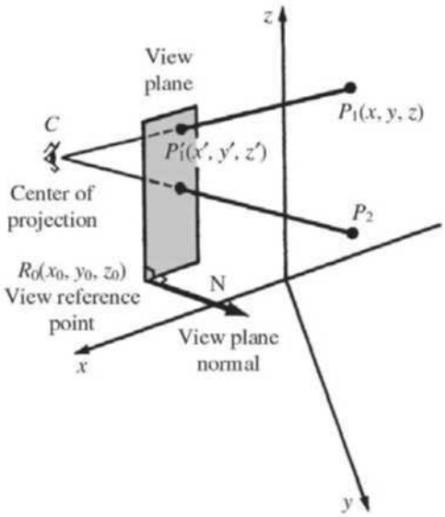


Fig. 7-3

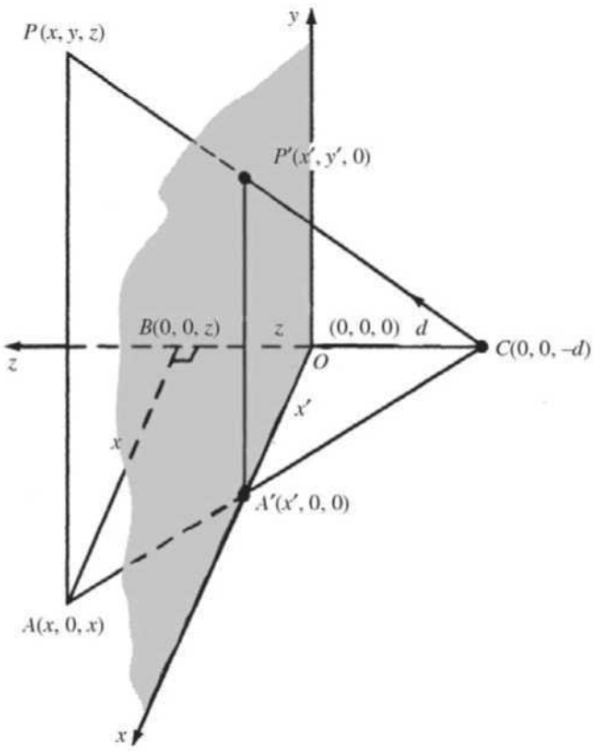


Fig. 7-4

EXAMPLE 1. The standard perspective projection is shown in Fig. 7-4. Here, the view plane is the xy plane, and the center of projection is taken as the point $C(0, 0, -d)$ on the negative z axis.

Using similar triangles ABC and $A'OC$, we find

$$x' = \frac{d \cdot x}{z + d} \qquad y' = \frac{d \cdot y}{z + d} \qquad z' = 0$$

The perspective transformation between object and image point is nonlinear and so cannot be represented as a 3×3 matrix transformation. However, if we use homogeneous coordinates, the perspective transformation can be represented as a 4×4 matrix:

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} d \cdot x \\ d \cdot y \\ 0 \\ z + d \end{pmatrix} = \begin{pmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & d \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

The general form of a perspective transformation is developed in Prob. 7.5.

Here the view plane is z = 0 (the xy-plane) and the COP is at C(0, 0, −d).

Step 1 — Ray from C(0, 0, −d) through P(x, y, z):

```
Point on ray = C + t(P − C) = (tx,  ty,  −d + t(z+d))
```

Step 2 — Find t when z-coordinate = 0 (hits the xy-plane):

```
−d + t(z + d) = 0   →   t = d / (z + d)
```

Step 3 — Projected coordinates:

```
x′ = tx = dx / (z + d)
y′ = ty = dy / (z + d)
z′ = 0
```

Result:

```
x′ = dx/(z+d),   y′ = dy/(z+d),   z′ = 0
```

4.5 Why a 3×3 Matrix Cannot Work — Homogeneous Coordinates

The problem:

From the general perspective result:

```
x′ = Dx / (n₁x + n₂y + n₃z)
```

The denominator $(n_1x + n_2y + n_3z)$ contains the input variables x, y, z — we are **dividing by a function of the input**. That is a **non-linear** operation.

A 3×3 matrix can only represent **linear transformations** (rotation, scaling, shearing). It cannot handle this division. So perspective projection cannot be expressed as a 3×3 matrix multiplication.

Without homogeneous coords	With homogeneous coords
Non-linear — no matrix possible	Linear 4×4 matrix + one final division
Cannot combine with other transforms	Transforms chain by matrix multiplication
Not GPU-friendly	Standard in graphics hardware pipelines

The fix — Homogeneous Coordinates:

Extend (x, y, z) to (x, y, z, 1) by adding an extra coordinate.

The 4×4 matrix encodes the numerators in the top rows and the denominator expression in the bottom row:

```
| D   0   0   0 |
| 0   D   0   0 |
| 0   0   D   0 |
| n₁  n₂  n₃  0 |
```

After multiplying by (x, y, z, 1), the result is (Dx, Dy, Dz, $n_1x+n_2y+n_3z$).

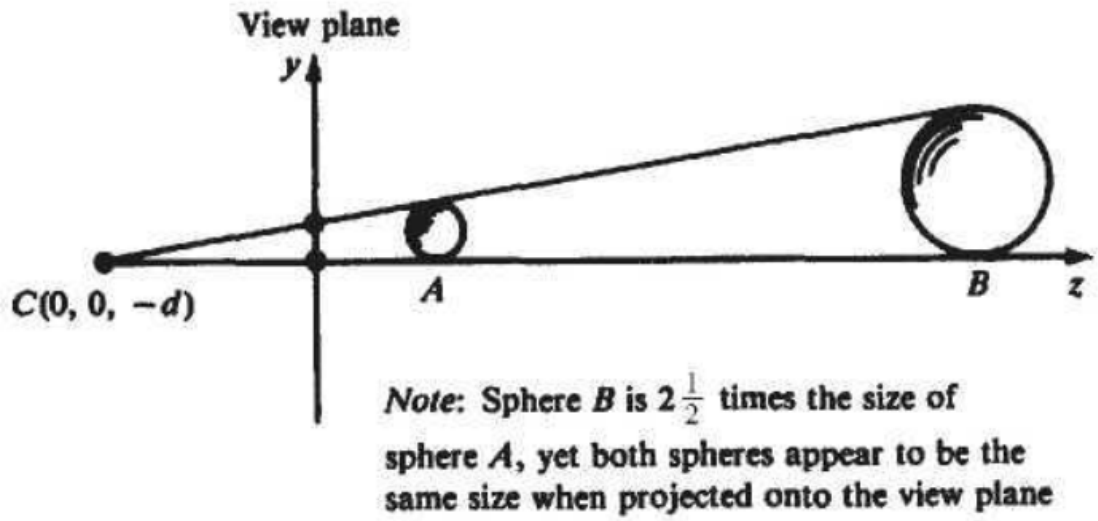
Perspective divide: divide by the last component to get x′, y′, z′.

> The division happens *after* the matrix multiplication — not inside it. This is the key trick that makes a non-linear operation GPU-compatible.

4.6 Perspective Anomalies

The perspective process introduces **four distinct visual anomalies**:

1. Perspective Foreshortening
The farther an object is from the center of projection, the smaller it appears (i.e. its projected size becomes smaller)



2. Vanishing Points
Sets of parallel 3D lines that are **not parallel to the view plane** appear to converge at a single point on the view plane called a vanishing point.

- Imagine a flat window (the **view plane**) in front of you.
- **Lines that are parallel to the window** (the view plane) stay parallel when you look at them—they **don't converge**.
- **Lines that are slanted away from the window** (not parallel to it) **appear to come together** at some point in the distance—that point is the **vanishing point**.

Example: Railway tracks appear to meet at the horizon because they are not parallel to your eyes' view plane—they go away from you into depth.?

- **Principal vanishing points** are formed by lines parallel to the x, y, or z axes.
- The number of principal vanishing points equals the number of principal axes the view plane intersects (gives 1-point, 2-point, or 3-point perspective).

3. View Confusion

Points **behind the COP** get projected upside-down and backward onto the view plane.(like mirror) This is why 3D clipping must happen *before* applying the perspective transformation.

EXAMPLE 2. For the standard perspective projection, the projections L_1' and L_2' of parallel lines L_1 and L_2 having the direction of the vector $\mathbf{K}$ appear to meet at the origin (Prob. 7.8). Refer to Fig. 7-6.

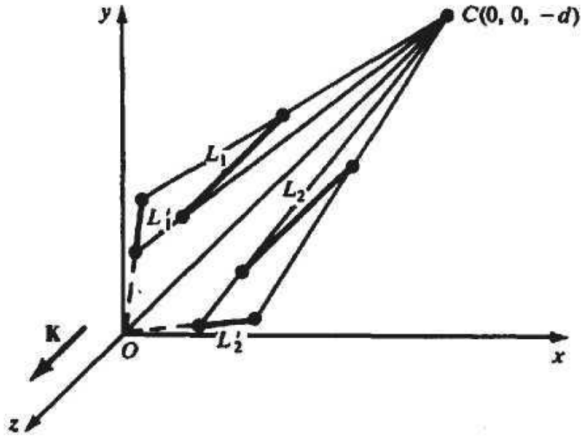


Fig. 7-6

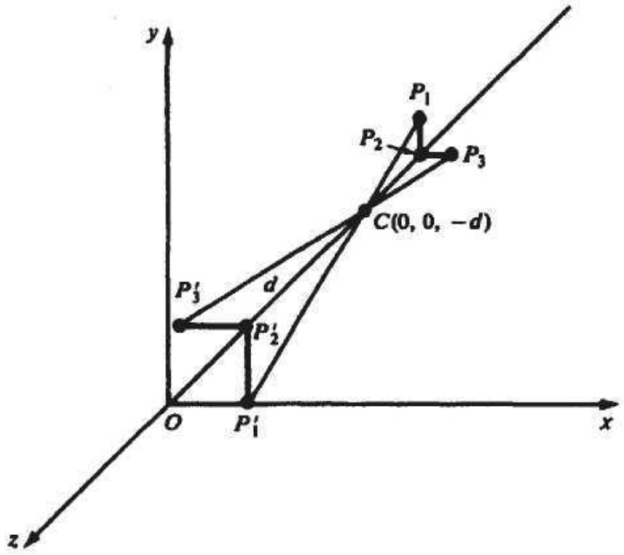


Fig. 7-7

4. Topological Distortion

The plane passing through the COP and parallel to the view plane is a singularity. Any point on it projects to infinity. A finite line segment crossing this plane is torn into two disconnected, infinitely long halves.

- Think of this like trying to take a photo of something right at the lens—it's too close to the eye, so it "blows up" in the image.

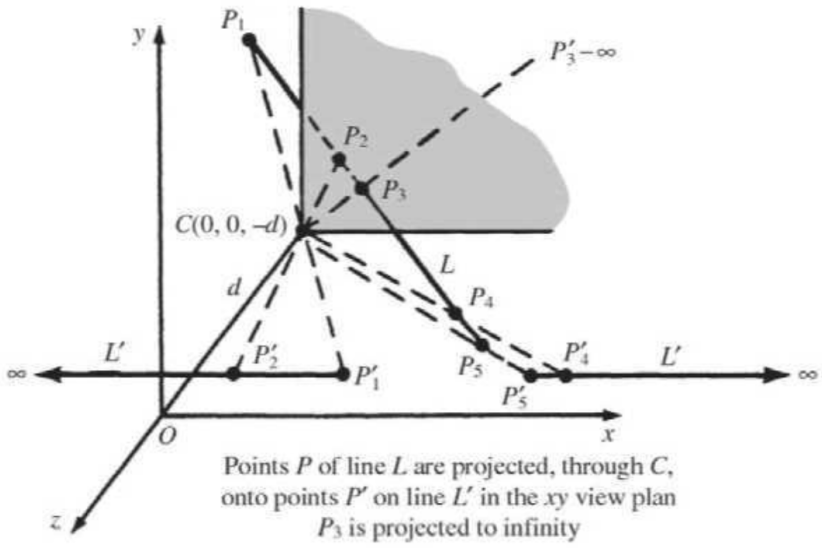


Fig. 7-8

---

5. Parallel Projection

In **parallel projection**, all projectors (lines along which points are projected) are **parallel to each other**, unlike perspective projection where they converge at a Center of Projection (COP). Conceptually, the COP is at infinity.

- **Direction of projection**: A fixed vector **V = (a, b, c)** governs the direction along which all points are projected.
- **Key property**: Preserves **true dimensions, shapes, and parallel relationships**, making it widely used in engineering and technical drawings.

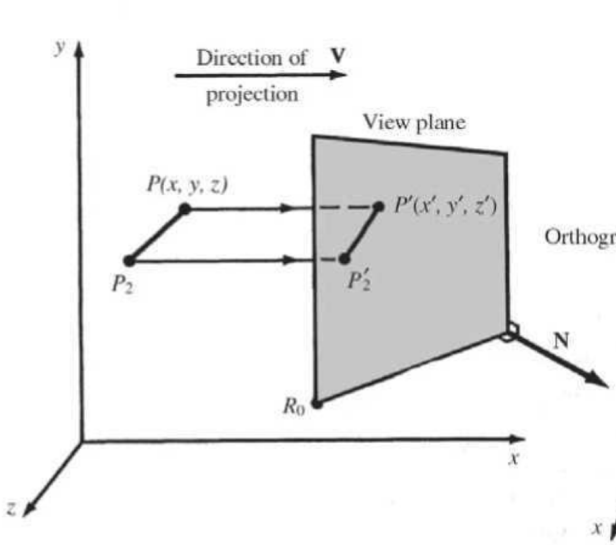


Fig. 7-9

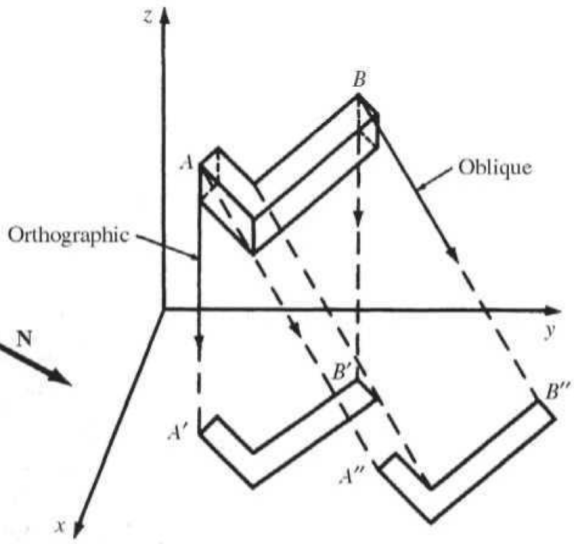


Fig. 7-10

5.1 Reference Plane and Projection Vector

- **Reference Plane**: Defined by a known plane equation (like z = 0 for xy-plane).
- **Projection vector**: **V = (a, b, c)** specifies the direction of projection.

Think of it as “pushing” each point along the same vector until it hits the view plane.

5.2 General Parallel Projection onto xy-Plane

Goal: Project a point **P(x, y, z)** onto the **xy-plane** (z = 0) along direction **V = (a, b, c)**.

7.10 Derive the equations of parallel projection onto the xy plane in the direction of projection $\mathbf{V} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$.

SOLUTION

From Fig. 7-22 we see that the vectors $\mathbf{V}$ and $\overline{\mathbf{PP'}}$ have the same direction. This means that $\overline{\mathbf{PP'}} = k\mathbf{V}$. Comparing components, we see that

$$x' - x = ka \qquad y' - y = kb \qquad z' - z = kc$$

So

$$k = -\frac{z}{c} \qquad x' = x - \frac{a}{c}z \qquad \text{and} \qquad y' = y - \frac{b}{c}z$$

In 3×3 matrix form, this is

$$Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & -\frac{a}{c} \\ 0 & 1 & -\frac{b}{c} \\ 0 & 0 & 0 \end{pmatrix}$$

and so $P' = Par_{\mathbf{V}} \cdot P$.

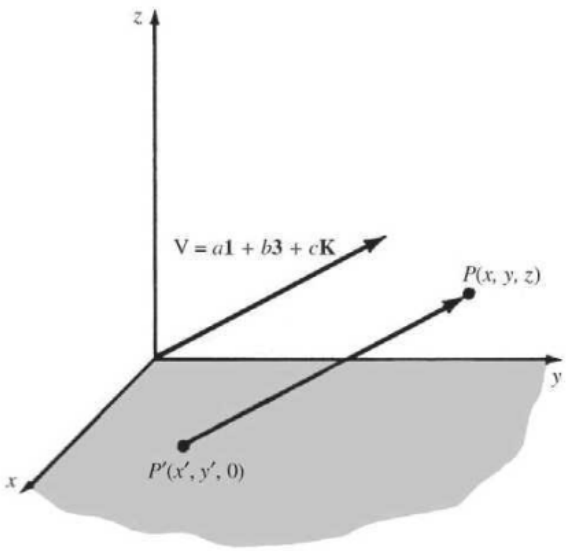


Fig. 7-22

Step 1 — Parametric form of the line

The line passing through **P** in direction **V**:

(x', y', z') = (x + a t, y + b t, z + c t)

where **t** is a scalar parameter.

Step 2 — Apply plane condition

The projected point lies on the plane **z′ = 0**:

0 = z + c t ⇒ t = -z/c

Step 3 — Solve for projected coordinates

```
x' = x + a t = x − (a/c) z
y' = y + b t = y − (b/c) z
z' = 0
```

Result:

```
x' = x − (a/c) z,   y' = y − (b/c) z,   z' = 0
```

Matrix form (3×3):

```
| 1   0   −a/c |   | x |   = | x − (a/c)z |
| 0   1   −b/c | * | y |   = | y − (b/c)z |
| 0   0    0   |   | z |   = |     0      |
```

> Linear transformation — no perspective divide is needed. This makes parallel projection simpler than perspective projection computationally.

5.3 Orthographic Projection (Special Case)

A special case of parallel projection:

- **Projection vector**: **V = (0, 0, 1)** (perpendicular to xy-plane).
- Substituting into the general formula:

```
x' = x,   y' = y,   z' = 0
```


Matrix form (4×4 homogeneous coordinates):

```
| 1   0   0   0 |
| 0   1   0   0 |
| 0   0   0   0 |
| 0   0   0   1 |
```

> Simply drops the z-coordinate. The simplest projection, widely used in CAD drawings.

5.3 Oblique Projection

Oblique projection is a parallel projection where V is **not perpendicular** to the view plane (not parallel to the normal N).

The derivation is identical to Section 5.1. The distinction is:

- V is parallel to N → **Orthographic**
- V is not parallel to N → **Oblique**

Two standard oblique subtypes (view plane = xy-plane):

Type	Lines perpendicular to view plane	Effective angle
Cavalier	Drawn at **full (1:1) length** — no foreshortening	~45°
Cabinet	Drawn at **half (1:2) length** — foreshortened 50%	~63.4° (arctan 2)

Cabinet looks more natural because the human eye perceives full-length receding lines as unrealistically long.

7.12 Find the general form of an oblique projection onto the xy plane.

SOLUTION

Refer to Fig. 7-24. Oblique projections (to the xy plane) can be specified by a number f and an angle θ. The number f prescribes the ratio that any line L perpendicular to the xy plane will be foreshortened after projection. The angle θ is the angle that the projection of any line perpendicular to the xy plane makes with the (positive) x axis.

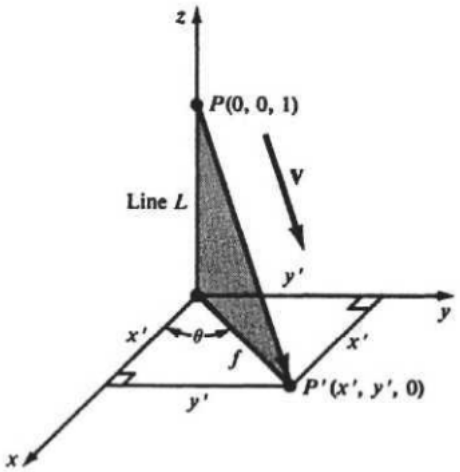


Fig. 7-24

To determine the projection transformation, we need to find the direction vector $\mathbf{V}$. From Fig. 7-24, with line L of length 1, we see that the vector $\overline{P'P}$ has the same direction as $\mathbf{V}$. We choose $\mathbf{V}$ to be this vector:

$$\mathbf{V} = \overline{P'P} = x'\mathbf{I} + y'\mathbf{J} - \mathbf{K} \qquad (= a\mathbf{I} + b\mathbf{J} + c\mathbf{K})$$

From Fig. 7-24 we find $a = x' = f\cos\theta$, $b = y' = f\sin\theta$, and $c = -1$.

From Prob. 7.10, the required transformation is

$$Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & f\cos\theta & 0 \\ 0 & 1 & f\sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

6. Named Projection Types — Quick Definitions

Name	Family	Key Characteristic
Isometric	Axonometric Orthographic	Equal angles with all 3 principal axes. All 3 axes foreshortened equally (~0.816). Projected angles between axes = 120°.

Name	Family	Key Characteristic
Dimetric	Axonometric Orthographic	Equal angles with exactly 2 of the 3 axes. Those 2 share one foreshortening ratio; the third is different.
Trimetric	Axonometric Orthographic	Unequal angles with all 3 axes. All three foreshortening ratios are different.
Cavalier	Oblique Parallel	Lines perpendicular to xy-plane drawn at full length. No foreshortening.
Cabinet	Oblique Parallel	Lines perpendicular to xy-plane drawn at half their actual length.

7. Exam Question Bank with Full Answers

Q1. Define view plane and center of projection for standard perspective projection.

(Dec 2018, Q1.d)

View Plane: The 2D surface onto which the 3D scene is projected to form the image.

Center of Projection (COP): The fixed 3D point from which all projection rays originate — like the position of the eye or camera. In the standard setup, the COP is at the origin (0, 0, 0) and the view plane is z = d.

Q2. What are the main differences between perspective and parallel projection?

(Dec 2018, Q6.b.i)

	Perspective	Parallel
COP location	Finite point	At infinity
Projectors	Converge at COP	All parallel
Realism	High	Low
Foreshortening	Yes	No
Parallel lines	Converge at vanishing points	Stay parallel
Use case	Rendering/animation	Engineering/architectural drawings

Q3. While dealing with perspective projection, which components must always be given? Why?

(Dec 2019, Q1.f)

You must always have:

1. The **Center of Projection (COP)** — where projection rays originate.
2. The **View Plane** — defined by either its equation (e.g., z = d) or a reference point R_0 and a normal vector N.

Why: Perspective projection finds P′ as the intersection of (a) a ray from the COP through P, and (b) the view plane. Without knowing where the ray starts (COP) and what surface it hits (view plane), the intersection cannot be computed.

Q4. Using the origin as the COP, derive the perspective transformation onto the plane through $R_0(x_0, y_0, z_0)$ with normal $N = n_1i + n_2j + n_3k$.

(Dec 2018, Q5.f / Dec 2019, Q6.c)

Step 1 — Plane equation:

$$N \cdot (P' - R_0) = 0 \quad \rightarrow \quad n_1x' + n_2y' + n_3z' = D$$
$$\text{where } D = n_1x_0 + n_2y_0 + n_3z_0$$

Step 2 — Ray from origin through P:

$$x' = \alpha x, \quad y' = \alpha y, \quad z' = \alpha z$$

Step 3 — Substitute into plane equation:

$$\alpha(n_1x + n_2y + n_3z) = D \quad \rightarrow \quad \alpha = D / (n_1x + n_2y + n_3z)$$

Step 4 — Final transformation:

$$x' = Dx / (n_1x + n_2y + n_3z)$$
$$y' = Dy / (n_1x + n_2y + n_3z)$$
$$z' = Dz / (n_1x + n_2y + n_3z)$$

Step 5 — Why not a 3×3 matrix: The denominator contains x, y, z — division by input variables is non-linear. A 3×3 matrix only handles linear operations.

Step 6 — Homogeneous 4×4 matrix:

$$\begin{vmatrix} D & 0 & 0 & 0 \\ 0 & D & 0 & 0 \\ 0 & 0 & D & 0 \\ n_1 & n_2 & n_3 & 0 \end{vmatrix}$$

Multiply by (x, y, z, 1) → get (Dx, Dy, Dz, $n_1x+n_2y+n_3z$). Divide by last component → gives x′, y′, z′ above.

Q5. Find the perspective projection onto the view plane z = d with COP at origin.

(Dec 2018, Q4.k / Dec 2021, Q1.e)

Special case of Q4 with N = (0, 0, 1) and R_o = (0, 0, d).

Compute D:

```
D = (0)(0) + (0)(0) + (1)(d) = d
```

Apply general formula:

```
x′ = dx / (0·x + 0·y + 1·z) = dx/z
y′ = dy/z
z′ = d
```

Matrix:

```
| d   0   0   0 |
| 0   d   0   0 |
| 0   0   0   0 |
| 0   0   1   0 |
```

Apply to (x, y, z, 1) → (dx, dy, 0, z). Divide by z → x′ = dx/z, y′ = dy/z.

Q6. Under the standard perspective transformation, find: (a) projected image of a point in z = −d, and (b) projected image of segment from P_1(−1, 1, −2d) to P_2(2, −2, 0).

(Dec 2018, Q2.d / Q6.a)

Using x′ = dx/z and y′ = dy/z from Q5:

(a) Point in the plane z = −d:

```
x′ = dx/(−d) = −x,   y′ = dy/(−d) = −y
```

The point (x, y, −d) projects to **(−x, −y, d)** — the image is inverted. This illustrates **view confusion** (point is behind the COP).

(b) Segment from $P_1(-1, 1, -2d)$ to $P_2(2, -2, 0)$:

Project P_1 ($z = -2d$):

```
x′ = d(−1)/(−2d) = 1/2,   y′ = d(1)/(−2d) = −1/2
```

Projected P_1 = **(1/2, −1/2, d)** ✓

Project P_2 ($z = 0$):

```
x′ = d(2)/0 = undefined (→ ∞)
y′ = d(−2)/0 = undefined (→ ∞)
```

P_2 lies on the plane $z = 0$ passing through the COP — its projection goes to infinity.

Result: The projected image is a **ray starting at (1/2, −1/2) on the view plane and extending to infinity**. This illustrates **topological distortion**.

Q7. Derive the equations of parallel projection onto the xy-plane in direction V = ai + bj + ck.

(Dec 2018, Q3.c.i)

Projector through P(x, y, z) parallel to V:

```
(x′, y′, z′) = (x + at,  y + bt,  z + ct)
```

From $z' = 0$:

```
0 = z + ct   →   t = −z/c
```

Substitute:

```
x′ = x − (a/c)z,   y′ = y − (b/c)z,   z′ = 0
```

Matrix form:

```
| 1   0   −a/c |   | x |   = | x − (a/c)z |
| 0   1   −b/c | * | y |   = | y − (b/c)z |
| 0   0    0   |   | z |   = |     0      |
```

Q8. Derive the general form of an oblique projection onto the xy-plane.

(Dec 2019, Q4.e)

7.12 Find the general form of an oblique projection onto the xy plane.

SOLUTION

Refer to Fig. 7-24. Oblique projections (to the xy plane) can be specified by a number f and an angle θ. The number f prescribes the ratio that any line L perpendicular to the xy plane will be foreshortened after projection. The angle θ is the angle that the projection of any line perpendicular to the xy plane makes with the (positive) x axis.

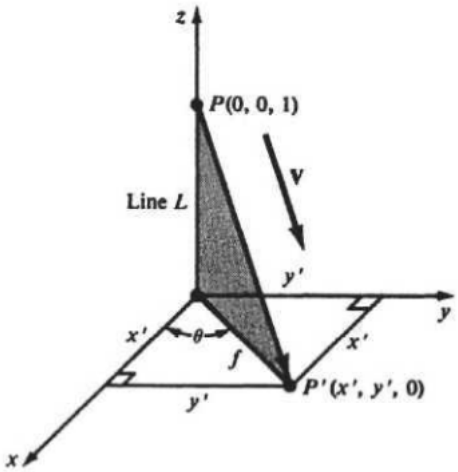


Fig. 7-24

To determine the projection transformation, we need to find the direction vector $\mathbf{V}$. From Fig. 7-24, with line L of length 1, we see that the vector $\overline{P'P}$ has the same direction as $\mathbf{V}$. We choose $\mathbf{V}$ to be this vector:

$$\mathbf{V} = \overline{P'P} = x'\mathbf{I} + y'\mathbf{J} - \mathbf{K} \qquad (= a\mathbf{I} + b\mathbf{J} + c\mathbf{K})$$

From Fig. 7-24 we find $a = x' = f\cos\theta$, $b = y' = f\sin\theta$, and $c = -1$.

From Prob. 7.10, the required transformation is

$$Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & f\cos\theta & 0 \\ 0 & 1 & f\sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Oblique projection is a parallel projection where V is not perpendicular to the xy-plane. Since the derivation is identical to Q7 (same direction-of-projection setup), the result is the same:

```
x′ = x − (a/c)z,   y′ = y − (b/c)z,   z′ = 0
```

The condition for oblique (not orthographic) is that at least one of (a, b) is nonzero — meaning V has a component along the plane, not just perpendicular to it.

Q9. What are the anomalies of perspective projection?

(Dec 2019, Q2.f / Dec 2021, Q6.f)

1. **Perspective foreshortening** — objects appear smaller with increasing distance from the COP.
2. **Vanishing points** — parallel 3D lines not parallel to the view plane appear to converge on the view plane.
3. **View confusion** — points behind the COP project upside-down and backward onto the view plane.
4. **Topological distortion** — a line crossing the plane through the COP parallel to the view plane is torn into two disconnected infinite halves.

Q10. What is a vanishing point?

(Dec 2019, Q4.c)

A vanishing point is a point on the view plane where the 2D projections of a set of mutually parallel 3D lines appear to converge. For example, parallel railway tracks appear to meet at the horizon.

Q11. Define dimetric projection.

(Dec 2020 Set-01 & Set-02, Q1.e)

Dimetric projection is an axonometric (orthographic parallel) projection where the direction of projection makes **equal angles with exactly two** of the three principal axes. Those two axes are foreshortened by the same ratio; the third axis has a different foreshortening ratio.

Q12. What is cabinet projection?

(Dec 2020 Set-02, Q1.e)

Cabinet projection is an oblique parallel projection where lines perpendicular to the projection plane are drawn at **half (1/2) of their actual length**. The projection rays make an angle of arctan(2) ≈ 63.4° with the view plane. This gives a more natural appearance than cavalier projection.

Q13. What is isometric projection?

(Dec 2021, Q5.h)

Isometric projection is an axonometric (orthographic parallel) projection where the direction of projection makes **equal angles with all three** principal axes (x, y, z). All three axes foreshorten equally by a ratio of approximately 0.816. The projected angles between the three axes are all 120°.
